



RV College of Engineering®

**HANDBOOK OF MATHEMATICS**  
**FOR**  
**SECOND YEAR B.E. PROGRAM**

## TRIGONOMETRY

### Basic Functions

- $\sin \theta = \frac{\text{Opposite Side}}{\text{Hypotenuse}}$
- $\sec \theta = \frac{\text{Hypotenuse}}{\text{Adjacent Side}} = \frac{1}{\cos \theta}$
- $\cos \theta = \frac{\text{Adjacent Side}}{\text{Hypotenuse}}$
- $\operatorname{cosec} \theta = \frac{\text{Hypotenuse}}{\text{Opposite Side}} = \frac{1}{\sin \theta}$
- $\tan \theta = \frac{\text{Opposite Side}}{\text{Adjacent Side}} = \frac{\sin \theta}{\cos \theta}$
- $\cot \theta = \frac{\text{Adjacent Side}}{\text{Opposite Side}} = \frac{1}{\tan \theta} = \frac{\cos \theta}{\sin \theta}$

### Identities

- $\sin(-x) = -\sin x$
- $\sin\left(\frac{\pi}{2} - x\right) = \cos x$
- $\sin\left(\frac{\pi}{2} + x\right) = \cos x$
- $\cos(-x) = \cos x$
- $\cos\left(\frac{\pi}{2} - x\right) = \sin x$
- $\cos\left(\frac{\pi}{2} + x\right) = -\sin x$
- $\tan(-x) = -\tan x$
- $\tan\left(\frac{\pi}{2} - x\right) = \cot x$
- $\tan\left(\frac{\pi}{2} + x\right) = -\cot x$
- $\sin(\pi - x) = \sin x$
- $\sin(\pi + x) = -\sin x$
- $\sin\left(\frac{3\pi}{2} - x\right) = -\cos x$
- $\cos(\pi - x) = -\cos x$
- $\cos(\pi + x) = -\cos x$
- $\cos\left(\frac{3\pi}{2} - x\right) = -\sin x$
- $\tan(\pi - x) = -\tan x$
- $\tan(\pi + x) = \tan x$
- $\tan\left(\frac{3\pi}{2} - x\right) = \cot x$
- $\sin\left(\frac{3\pi}{2} + x\right) = -\cos x$
- $\sin(x + y) = \sin x \cos y + \cos x \sin y$
- $\cos\left(\frac{3\pi}{2} + x\right) = \sin x$
- $\cos(x + y) = \cos x \cos y - \sin x \sin y$
- $\tan\left(\frac{3\pi}{2} + x\right) = -\cot x$
- $\tan(x + y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}$
- $\sin(2x) = 2 \sin x \cos x$
- $\cos^2 x + \sin^2 x = 1$
- $\cos(2x) = \cos^2 x - \sin^2 x = 1 - 2 \sin^2 x$
- $\sec^2 x - \tan^2 x = 1$
- $\quad \quad \quad = 2 \cos^2 x - 1$
- $\operatorname{cosec}^2 x - \cot^2 x = 1$
- $\tan(2x) = \frac{2 \tan x}{1 - \tan^2 x}$
- $\sin 3x = 3 \sin x - 4 \sin^3 x$
- $\sin x + \sin y = 2 \sin \frac{x+y}{2} \cos \frac{x-y}{2}$
- $\cos 3x = 4 \cos^3 x - 3 \cos x$
- $\sin x - \sin y = 2 \cos \frac{x+y}{2} \sin \frac{x-y}{2}$
- $\cos x + \cos y = 2 \cos \frac{x+y}{2} \cos \frac{x-y}{2}$
- $\cos x - \cos y = -2 \sin \frac{x+y}{2} \sin \frac{x-y}{2}$

## BASIC CALCULUS

### Differentiation

| $f(x)$                             | $f'(x)$                            | $f(x)$                         | $f'(x)$                      |
|------------------------------------|------------------------------------|--------------------------------|------------------------------|
| $a$                                | $0$                                | $a^x$                          | $a^x \log_e a$               |
| $x^n, n \neq -1$                   | $nx^{n-1}$                         | $e^{ax}$                       | $ae^{ax}$                    |
| $\sqrt{x}$                         | $\frac{1}{2\sqrt{x}}$              | $\log_e x$                     | $\frac{1}{x}$                |
| $\sin x$                           | $\cos x$                           | $\sin^{-1} x$                  | $\frac{1}{\sqrt{1-x^2}}$     |
| $\cos x$                           | $-\sin x$                          | $\cos^{-1} x$                  | $-\frac{1}{\sqrt{1-x^2}}$    |
| $\tan x$                           | $\sec^2 x$                         | $\tan^{-1} x$                  | $\frac{1}{1+x^2}$            |
| $\operatorname{cosec} x$           | $-\cot x \operatorname{cosec} x$   | $\operatorname{cosec}^{-1} x$  | $-\frac{1}{ x \sqrt{x^2-1}}$ |
| $\sec x$                           | $\tan x \sec x$                    | $\sec^{-1} x$                  | $\frac{1}{ x \sqrt{x^2-1}}$  |
| $\cot x$                           | $-\operatorname{cosec}^2 x$        | $\cot^{-1} x$                  | $-\frac{1}{1+x^2}$           |
| $\sinh x = \frac{e^x - e^{-x}}{2}$ | $\cosh x$                          | $\sinh^{-1} x$                 | $\frac{1}{\sqrt{x^2+1}}$     |
| $\cosh x = \frac{e^x + e^{-x}}{2}$ | $\sinh x$                          | $\cosh^{-1} x$                 | $\frac{1}{\sqrt{x^2-1}}$     |
| $\tanh x$                          | $\operatorname{sech}^2 x$          | $\tanh^{-1} x$                 | $\frac{1}{1-x^2}$            |
| $\operatorname{cosech} x$          | $-\coth x \operatorname{cosech} x$ | $\operatorname{cosech}^{-1} x$ | $-\frac{1}{ x \sqrt{x^2+1}}$ |
| $\operatorname{sech} x$            | $-\tanh x \operatorname{sech} x$   | $\operatorname{sech}^{-1} x$   | $-\frac{1}{ x \sqrt{1-x^2}}$ |
| $\coth x$                          | $-\operatorname{cosech}^2 x$       | $\coth^{-1} x$                 | $\frac{1}{1-x^2}$            |

### Rules of differentiation

- $\frac{d}{dx}(fg) = gf' + fg'$

- $\frac{d}{dx}\left(\frac{f}{g}\right) = \frac{gf' - fg'}{g^2}$
- $\frac{d}{dx}(f(t)) = \frac{d}{dt}(f(t)) \frac{dt}{dx}$

## Integration

| $f(x)$                       | $\int f(x)dx$                                                                          | $f(x)$                       | $\int f(x)dx$                                      |
|------------------------------|----------------------------------------------------------------------------------------|------------------------------|----------------------------------------------------|
| $x^n$                        | $\frac{x^{n+1}}{n+1}$                                                                  | $\frac{1}{x}$                | $\log_e x$                                         |
| $e^{ax}$                     | $\frac{e^{ax}}{a}$                                                                     | $\log_e x$                   | $x(\log_e x - 1)$                                  |
| $a^x$                        | $\frac{a^x}{\log_e a}$                                                                 | $\operatorname{cosec} x$     | $\log_e(\operatorname{cosec} x - \cot x)$          |
| $\sin x$                     | $-\cos x$                                                                              | $\sec x$                     | $\log_e(\sec x + \tan x)$                          |
| $\cos x$                     | $\sin x$                                                                               | $\cot x$                     | $\log_e \sin x$                                    |
| $\tan x$                     | $\log_e \sec x$                                                                        | $\sec^2 x$                   | $\tan x$                                           |
| $\sinh x$                    | $\cosh x$                                                                              | $\operatorname{cosec}^2 x$   | $-\cot x$                                          |
| $\cosh x$                    | $\sinh x$                                                                              | $\tanh x$                    | $\log_e \cosh x$                                   |
| $\frac{1}{\sqrt{a^2 - x^2}}$ | $\sin^{-1}\left(\frac{x}{a}\right)$                                                    | $\frac{1}{\sqrt{a^2 + x^2}}$ | $\sinh^{-1}\left(\frac{x}{a}\right)$               |
| $\frac{1}{a^2 + x^2}$        | $\frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right)$                                        | $\frac{1}{\sqrt{x^2 - a^2}}$ | $\cosh^{-1}\left(\frac{x}{a}\right)$               |
| $\frac{1}{a^2 - x^2}$        | $\frac{1}{2a} \log_e \left(\frac{a+x}{a-x}\right)$                                     | $\frac{1}{x^2 - a^2}$        | $\frac{1}{2a} \log_e \left(\frac{x-a}{x+a}\right)$ |
| $\sqrt{a^2 - x^2}$           | $\frac{1}{2} \left[ x\sqrt{a^2 - x^2} + a^2 \sin^{-1}\left(\frac{x}{a}\right) \right]$ | $e^{ax} \sin bx$             | $\frac{e^{ax}}{a^2 + b^2} (a \sin bx - b \cos bx)$ |
| $u(x)v(x)$                   | $u \int v dx - \int \left[ \frac{du}{dx} \left[ \int v dx \right] dx \right]$          | $e^{ax} \cos bx$             | $\frac{e^{ax}}{a^2 + b^2} (a \cos bx + b \sin bx)$ |

## COMPLEX ANALYSIS

### Algebra of complex numbers:

- If  $z = x + iy$ , then  $|z| = \sqrt{x^2 + y^2}$  is non-negative real number.
- $z = x + iy$  is represented by a point  $P(x, y)$  in the  $XY$  plane,  $x$  - axis is real axis,  $y$  - axis is imaginary axis, plane is complex plane.
- $z = re^{i\theta}$  is the polar form of complex number  $z$  where  $r = \sqrt{x^2 + y^2}$  and  $\theta = \tan^{-1}\left(\frac{y}{x}\right)$ .
- $|z_1 - z_2| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$  represents the distance between the points  $z_1$  and  $z_2$  in complex plane.
- $|z - z_0| = R$  represents complex equation of circle with centre  $z_0$  and radius  $R$ .
- $|z - z_0| < R$  represents the region with in, but not on, a circle of radius  $R$  centred at the point  $z_0$ , the point  $z_0$  is said to be interior point.
- $|z - z_0| \leq R$  represents the region with in, and on, a circle of radius  $R$  centred at the point  $z_0$ .
- $|z - z_0| > R$  represents the region outside the circle with centre  $z_0$  and radius  $R$ .

### Cauchy-Riemann (C-R) equations:

- In Cartesian form: If  $f(z) = u(x, y) + i v(x, y)$  is differentiable at  $z = x + iy$ , then  $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$  &  $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$
- In polar form: If  $f(z) = u(r, \theta) + i v(r, \theta)$ , then  $\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta}$  and  $\frac{\partial v}{\partial r} = -\frac{1}{r} \frac{\partial u}{\partial \theta}$

**Harmonic Functions:** A function  $\phi$  is said to be a harmonic function if it satisfies Laplace equation  $\nabla^2 \phi = 0$ .

- In Cartesian form  $\phi(x, y)$  is harmonic if  $\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0$ .
- In polar form  $\phi(r, \theta)$  is harmonic if  $\frac{\partial^2 \phi}{\partial r^2} + \frac{1}{r} \frac{\partial \phi}{\partial r} + \frac{1}{r^2} \frac{\partial^2 \phi}{\partial \theta^2} = 0$ .

**Taylor theorem:** Taylor series expansion for the function  $f(z)$  about the point  $z = a$  is

$$f(z) = f(a) + (z - a)f'(a) + (z - a)^2 \frac{f''(a)}{2!} + \dots$$

**Maclaurin theorem:** Maclaurin series for the function  $f(z)$  about the point  $z = 0$  is

$$f(z) = f(0) + zf'(0) + z^2 \frac{f''(0)}{2!} + \dots$$

**Binomial Expansion:** If  $|x| < 1$ , then

- $(1 + x)^{-1} = 1 - x + x^2 - x^3 + \dots$
- $(1 - x)^{-1} = 1 + x + x^2 + x^3 + \dots$
- $(1 - x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \dots$
- $(1 + x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + \dots$

**Laurent's theorem:** Let  $c_1$  and  $c_2$  are concentric circles centered at " $a$ ", then Laurent's series of  $f(z)$  about the point  $z = a$ ,

$$f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n + \sum_{n=1}^{\infty} \frac{b_n}{(z-a)^n},$$

$$\text{where } a_n = \frac{1}{2\pi i} \int_C \frac{f(w)}{(w-a)^{n+1}} dw \quad b_n = \frac{1}{2\pi i} \int_C \frac{f(w)}{(w-a)^{-n+1}} dw \quad \text{for } n = 0, 1, 2 \dots$$

**Determination of poles:** If  $f(z) = \frac{\phi(z)}{(z-a)^m}$  where  $\phi(z)$  is analytic and not zero at the point ' $a$ ', then ' $a$ ' is a pole of order  $m$  of  $f(z)$ . The poles of  $f(z)$  may be obtained by solving the equation  $\frac{1}{f(z)} = 0$ .

**Residue:** The coefficient of  $\frac{1}{z-a}$  in the Laurent's expansion of  $f(z)$  is the Residue of  $f(z)$  at the pole  $z = a$ .

**Determination of a residue:** If ' $a$ ' is a pole of order  $m \geq 1$  of  $f(z)$ , then residue of  $f(z)$  at ' $a$ ' is  $\frac{1}{(m-1)!} \left[ \lim_{z \rightarrow a} \frac{d^{m-1}}{dz^{m-1}} \{(z-a)^m f(z)\} \right]$ .

**Cauchy's residue theorem:** Let  $C$  be a simple closed curve and  $f(z)$  be analytic within and on  $C$  except at a finite number of poles  $a_1, a_2, \dots, a_n$  which lie inside  $C$ , then

$$\int_C f(z) dz = 2\pi i (R_1 + R_2 + \dots + R_n)$$

where  $R_1, R_2 \dots R_n$  are the residues of  $f(z)$  at  $a_1, a_2, \dots, a_n$  respectively.

## PARTIAL DIFFERENTIAL EQUATIONS

**Lagrange's linear equation:** The first order linear partial differential equation of the form  $P \frac{\partial u}{\partial x} + Q \frac{\partial u}{\partial y} = R$ , where  $P, Q$  and  $R$  are functions of  $x, y, z$  is known as Lagrange's Linear equation.

**Subsidiary/Auxiliary equation:** The equation  $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$  is known as the subsidiary/auxiliary equation of as Lagrange's Linear equation  $P \frac{\partial u}{\partial x} + Q \frac{\partial u}{\partial y} = R$ .

**One-dimensional wave equation:**  $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$ , where  $c^2 = \frac{T}{\rho}$  the phase speed,  $T$  is the tension, and  $\rho$  density of the string.

**One-dimensional heat equation:**  $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ , where  $c^2 = \frac{\kappa}{s\rho}$  the thermal diffusivity,  $\kappa$  thermal conductivity,  $s$  specific heat and  $\rho$  density of the material of the body.

**Two-dimensional Laplace equation:**  $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ .

## NUMERICAL SOLUTION OF PARTIAL DIFFERENTIAL EQUATIONS

**Laplace equation:**  $u_{xx} + u_{yy} = 0$

- Standard 5- point formula:  $u_{i,j} = \frac{1}{4} [u_{i-1,j} + u_{i+1,j} + u_{i,j-1} + u_{i,j+1}]$
- Diagonal 5- point formula:  $u_{i,j} = \frac{1}{4} [u_{i-1,j+1} + u_{i+1,j-1} + u_{i+1,j+1} + u_{i-1,j-1}]$ .

**One-dimensional heat equation:**  $u_t = c^2 u_{xx}$

- Schmidt formula:  $u_{i,j+1} = \alpha(u_{i-1,j} + u_{i+1,j}) + (1 - 2\alpha)u_{i,j}$  where  $\alpha = \frac{kc^2}{h^2}$ .
- Bendre-Schmidt relation:  $u_{i,j+1} = \frac{1}{2} [u_{i-1,j} + u_{i+1,j}]$  when  $\alpha = \frac{1}{2}$ .

**One-dimensional wave equation:**  $u_{tt} = c^2 u_{xx}$

- Explicit formula:  $u_{i,j+1} = \beta^2(u_{i+1,j} + u_{i-1,j}) + 2(1 - \beta^2)u_{i,j} - u_{i,j-1}$ , where  $\beta^2 = \frac{c^2 k^2}{h^2}$ .
- For  $\beta = 1$  and  $k = \frac{h}{c}$ ,  $u_{i,j+1} = u_{i-1,j} + u_{i+1,j} - u_{i,j-1}$ .

The above scheme is used with standard initial and boundary conditions.

## CALCULUS OF VARIATION

**Euler's equation:** A necessary condition for the functional  $I = \int_{x_1}^{x_2} f(x, y, y') dx$  to be an extremum is that  $\frac{\partial f}{\partial y} - \frac{d}{dx} \left( \frac{\partial f}{\partial y'} \right) = 0$ .

**Alternate forms:**

- $\frac{d}{dx} \left( f - y' \frac{\partial f}{\partial y'} \right) = \frac{\partial f}{\partial x}$
- $\frac{\partial f}{\partial x} - \frac{d}{dx} \left( f - y' \frac{\partial f}{\partial y'} \right) = 0$
- $\frac{\partial f}{\partial y} - \frac{\partial^2 f}{\partial x \partial y'} - \frac{\partial^2 f}{\partial y \partial y'} y' - \frac{\partial^2 f}{\partial (y')^2} y'' = 0$

**Cases of Euler's Equation:**

- If  $f$  is only the function of  $y'$ , then Euler's equation will be:  $\frac{df}{dy'} = c$  where  $c$  is an arbitrary constant.
- If  $f$  is independent of  $y$ , then Euler's equation will be:  $\frac{\partial f}{\partial y'} = c$  where  $c$  is an arbitrary constant.
- If  $f$  is independent of  $y'$  then Euler's equation will be:  $\frac{\partial f}{\partial y} = 0$ .
- If  $f$  is independent of  $x$  and  $y$  then Euler's equation will be:  $y'' \frac{\partial^2 f}{\partial y'^2} = 0$ .
- If  $f$  does not contain  $x$  then Euler's equation will be:  $f - y' \frac{\partial f}{\partial y'} = c$  where  $c$  is an arbitrary constant.

**Cartesian coordinate system:**  $u_1 = x, u_2 = y, u_3 = z$

Element of arc length  $ds = \sqrt{(dx)^2 + (dy)^2 + (dz)^2}$ .

**Cylindrical coordinate system:**  $u_1 = r, u_2 = \theta, u_3 = z$

Element of arc length  $ds = \sqrt{(dr)^2 + r^2(d\theta)^2 + (dz)^2}$

**Spherical coordinate system:**  $u_1 = r, u_2 = \theta$  and  $u_3 = \phi$ .

Element of arc length  $ds = \sqrt{(dr)^2 + r^2(d\theta)^2 + r^2 \sin^2 \theta (d\phi)^2}$ .

## LAPLACE TRANSFORM

### Gamma function:

- $\Gamma(n) = \int_0^\infty e^{-x} x^{n-1} dx, (n > 0)$
- $\Gamma(n) = 2 \int_0^\infty e^{-x^2} x^{2n-1} dx$
- $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$
- $\Gamma(1) = 1$
- $\Gamma(n+1) = \begin{cases} n\Gamma(n), & n > 0 \\ n!, & n \text{ positive integer} \end{cases}$
- $\Gamma(n) = \frac{\Gamma(n+1)}{n}, (n < 0, \neq -1, -2, \dots)$

### Beta Function:

- $\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$
- $\beta(m, n) = 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta$
- $\beta(m, n) = \int_0^\infty \frac{t^{m-1}}{(1+t)^{m+n}} dt$
- $\beta(m, n) = \beta(n, m)$
- $\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$

**Laplace transform of  $f(t)$ :**  $L\{f(t)\} = \int_0^\infty e^{-st} f(t) dt$

### Transform of elementary functions:

- $L(e^{at}) = \frac{1}{s-a}, s > a$
- $L(\sinh at) = L\left(\frac{e^{at}-e^{-at}}{2}\right) = \frac{a}{s^2-a^2}, s > |a|$
- $L(\sin at) = \frac{a}{s^2+a^2}, s > 0$
- $L(\cosh at) = \frac{s}{s^2-a^2}, s > |a|$
- $L(\cos at) = \frac{s}{s^2+a^2}, s > 0$
- $L(t^n) = \frac{\Gamma(n+1)}{s^{n+1}}$
- $L[H(t-a)] = \frac{e^{-as}}{s}$ , where  $H$  is Heaviside unit step function

### Properties of Laplace transform:

- $L[af(t) + bg(t)] = aL[f(t)] + bL[g(t)]$ .
- If  $L[f(t)] = F(s)$ , then  $L[f(at)] = \frac{1}{a}F\left(\frac{s}{a}\right)$ , where  $a$  is a positive constant.
- Let  $a$  be any real constant then  $L[e^{at}f(t)] = F(s-a)$
- If  $L[f(t)] = F(s)$ , then  $L[t^n f(t)] = (-1)^n \frac{d^n}{ds^n} F(s), n = 1, 2, 3, \dots$
- If  $L[f(t)] = F(s)$ , then  $L\left(\frac{f(t)}{t}\right) = \int_s^\infty F(s) ds$ .
- If  $L[f(t)] = F(s)$ , then  $L\{f^n(t)\} = s^n L\{f(t)\} - s^{n-1}f(0) - s^{n-2}f'(0) - \dots - f^{n-1}(0)$
- If  $L[f(t)] = F(s)$ , then  $L\int_0^t f(t) dt = \frac{1}{s}F(s)$
- If  $f(t)$  is a periodic function of period  $T$ , then  $L\{f(t)\} = \frac{1}{1-e^{-sT}} \int_0^T e^{-st} f(t) dt$ .

- If  $L\{f(t)\} = F(s)$ , then  $L[f(t-a)H(t-a)] = e^{-as}F(s)$
- If  $f(t)$  is a continuous function at  $t = a$ , then  $\int_0^\infty f(t)\delta(t-a)dt = f(a)$ , where  $\delta(t-a)$  is unit impulse function.

**Inverse Laplace transform of  $F(s)$  using Convolution theorem:** If  $L^{-1}[F(s)] = f(t)$  and  $L^{-1}[G(s)] = g(t)$ , then  $L^{-1}[F(s)G(s)] = \int_0^t f(u)g(t-u)du = f(t) * g(t)$ .

## FOURIER SERIES

**Fourier series of  $f(x)$  in the interval  $(a, a + 2l)$ :**

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right),$$

where Fourier coefficients  $a_0, a_n, b_n$  are given by

$$a_0 = \frac{1}{l} \int_a^{a+2l} f(x) dx, \quad a_n = \frac{1}{l} \int_a^{a+2l} f(x) \cos\left(\frac{n\pi x}{l}\right) dx, \quad n = 1, 2, 3, \dots \text{ and}$$

$$b_n = \frac{1}{l} \int_a^{a+2l} f(x) \sin\left(\frac{n\pi x}{l}\right) dx, \quad n = 1, 2, 3, \dots$$

**Complex Fourier Series of  $f(x)$  in the interval  $(a, a + 2l)$ :**

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{\frac{in\pi x}{l}}, \text{ where } c_n = \frac{1}{2l} \int_a^{a+2l} f(x) e^{-\frac{in\pi x}{l}} dx, n = 0, \pm 1, \pm 2, \pm 3, \dots$$

**Relation between Fourier and complex Fourier coefficients:**

$$a_0 = 2c_0, a_n = c_n + c_{-n}, b_n = i(c_n - c_{-n})$$

**Half-range Fourier series:**

- Sine series of  $f(x)$  in  $(0, l)$ :  $f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right)$ , where  $b_n = \frac{2}{l} \int_0^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx$ .
- Cosine series of  $f(x)$  in  $(0, l)$ :  $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right)$ , where  $a_0 = \frac{2}{l} \int_0^l f(x) dx$  and  $a_n = \frac{2}{l} \int_0^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx$

**Harmonic Analysis:** Let the periodic function  $y = f(x)$  takes values  $y_0, y_1, y_2, \dots, y_m$  corresponding to a given set of equi-spaced values  $x_0, x_1, x_2, \dots, x_m$  in  $(a, a + 2l)$ , Fourier series of  $f(x)$  is  $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right)$ , where  $a_0, a_n$  and  $b_n$  are computed using the formulae:

$$a_0 = 2 \frac{\sum y}{m}, \quad a_n = 2 \frac{\sum y \cos\left(\frac{n\pi x}{l}\right)}{m}, \quad b_n = 2 \frac{\sum y \sin\left(\frac{n\pi x}{l}\right)}{m}.$$

- Let the periodic function  $y = f(x)$  takes values  $y_0, y_1, y_2, \dots, y_m$  corresponding to a given set of equi spaced values  $x_0, x_1, x_2, \dots, x_m$  in  $(0, l)$ , Half range cosine Fourier series and Half range sine Fourier series of  $f(x)$  are respectively:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right)$$

$$= \frac{1}{2} a_0 + a_1 \cos\left(\frac{\pi x}{l}\right) + a_2 \cos\left(\frac{2\pi x}{l}\right) + \dots$$

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right)$$

$$= b_1 \sin\left(\frac{\pi x}{l}\right) + b_2 \sin\left(\frac{2\pi x}{l}\right) + \dots,$$

where  $a_0$ ,  $a_n$  and  $b_n$  are computed from the table by using the formulae:

$$a_0 = 2 \frac{\Sigma y}{m}, \quad a_n = 2 \frac{\Sigma y \cos\left(\frac{n\pi x}{l}\right)}{m}, \quad b_n = 2 \frac{\Sigma y \sin\left(\frac{n\pi x}{l}\right)}{m}$$

## FOURIER TRANSFORMS

### Complex Fourier transform or Fourier transform of $f(x)$ :

- Fourier transform of  $f(x)$ :  $\hat{f}(\alpha) = F[f(x)] = \int_{-\infty}^{\infty} f(x) e^{i\alpha x} dx$ , provided the integral exists.
- Inverse Fourier transform of  $\hat{f}(\alpha)$ :  $F^{-1}[\hat{f}(\alpha)] = f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(\alpha) e^{-i\alpha x} d\alpha$ .

### Fourier sine and cosine transforms:

- Fourier sine transform of  $f(x)$ :  $\hat{f}_s(\alpha) = F_s[f(x)] = \int_0^{\infty} f(x) \sin \alpha x \, dx$
- Inverse Fourier sine transform  $\hat{f}_s(\alpha)$ :  $F_s^{-1}[\hat{f}_s(\alpha)] = \frac{2}{\pi} \int_{-\infty}^{\infty} \hat{f}_s(\alpha) \sin \alpha x \, d\alpha$
- Fourier cosine transform of  $f(x)$ :  $\hat{f}_c(\alpha) = F_c[f(x)] = \int_0^{\infty} f(x) \cos \alpha x \, dx$
- Inverse Fourier cosine transform of  $\hat{f}_s(\alpha)$ :  $F_c^{-1}[\hat{f}_s(\alpha)] = \frac{2}{\pi} \int_{-\infty}^{\infty} \hat{f}_c(\alpha) \cos \alpha x \, d\alpha$

### Relation between Fourier sine and cosine transform:

- $F_s[xf(x)] = -\frac{d}{d\alpha} F_c[f(x)]$  and  $F_c[xf(x)] = \frac{d}{d\alpha} F_s[f(x)]$

### Properties of Fourier transforms:

- For any two functions  $f(x)$  and  $\phi(x)$  (whose Fourier transforms exist) and any two constants  $a$  and  $b$ ,  $F[af(x) + b\phi(x)] = aF[f(x)] + bF[\phi(x)]$
- If  $F[f(x)] = \hat{f}(\alpha)$ , then for any non-zero constant  $a$ ,  $F[f(ax)] = \frac{1}{|a|} \hat{f}\left(\frac{\alpha}{a}\right)$
- If  $F[f(x)] = \hat{f}(\alpha)$ , then for any non-zero constant  $a$ ,
  - (i)  $F[f(x - a)] = e^{i\alpha a} \hat{f}(\alpha)$
  - (ii)  $F[e^{i\alpha x} f(x)] = \hat{f}(\alpha + a)$
- $F[f'(x)] = -isF[f(x)]$  and  $F[f''(x)] = -s^2 F[f(x)]$
- If  $F[f(x)] = \hat{f}(\alpha)$ , then
  - (i)  $F[f(x) \cos ax] = \frac{1}{2} [\hat{f}(\alpha + a) + \hat{f}(\alpha - a)]$  and
  - (ii)  $F[f(x) \sin ax] = \frac{1}{2} [\hat{f}(\alpha + a) - \hat{f}(\alpha - a)]$ , where ' $a$ ' is a real constant.
- If  $\hat{f}_s(\alpha)$  and  $\hat{f}_c(\alpha)$  are Fourier sine and cosine transforms of  $f(x)$  respectively, then
  - (i)  $F_s[f(x) \cos ax] = \frac{1}{2} [\hat{f}_s(\alpha + a) + \hat{f}_s(\alpha - a)]$ ,
  - (ii)  $F_c[f(x) \sin ax] = \frac{1}{2} [\hat{f}_s(\alpha + a) - \hat{f}_s(\alpha - a)]$ ,

$$(iii) F_s[f(x) \sin ax] = \frac{1}{2} [\hat{f}_c(\alpha - a) + \hat{f}_c(\alpha + a)].$$

**Convolution theorem:** If  $F[f(x)] = \hat{f}(\alpha)$  and  $F[g(x)] = \hat{g}(\alpha)$ , then

$$F^{-1}[\hat{f}(\alpha)\hat{g}(\alpha)] = f(x) * g(x) = \int_{-\infty}^{\infty} f(u)g(x-u)du$$

**Parseval's identity:**

- If the Fourier transform of  $f(x)$  and  $g(x)$  are  $F(\alpha)$  and  $G(\alpha)$ , respectively, then
  - $\frac{1}{2\pi} \int_{-\infty}^{\infty} F(\alpha)\bar{G}(\alpha)d\alpha = \int_{-\infty}^{\infty} f(x)\bar{g}(x)dx$
  - $\frac{1}{2\pi} \int_{-\infty}^{\infty} F(\alpha)\bar{G}(\alpha)d\alpha = \int_{-\infty}^{\infty} |f(x)|^2 dx$
- If Fourier cosine and sine transform of  $f(x)$  and  $g(x)$  are  $F_c(\alpha)$ ,  $G_c(\alpha)$  and  $F_s(\alpha)$ ,  $G_s(\alpha)$  respectively, then
  - $\frac{2}{\pi} \int_0^{\infty} F_c(\alpha)G_c(\alpha)d\alpha = \int_0^{\infty} f(x)g(x)dx$
  - $\frac{2}{\pi} \int_0^{\infty} F_s(\alpha)G_s(\alpha)d\alpha = \int_0^{\infty} f(x)g(x)dx$
  - $\frac{2}{\pi} \int_0^{\infty} [F_c(\alpha)]^2 d\alpha = \int_0^{\infty} [f(x)]^2 dx$
  - $\frac{2}{\pi} \int_0^{\infty} [F_s(\alpha)]^2 d\alpha = \int_0^{\infty} [f(x)]^2 dx$

## LINEAR ALGEBRA

**Basic transformation matrices:**

- Stretch matrix/dilation matrix:  $\begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix}$ , where  $a$  is real constant.
- Rotation matrix:  $\begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$
- Projection matrix:  $\begin{bmatrix} \cos^2(\theta) & \sin(\theta)\cos(\theta) \\ \sin(\theta)\cos(\theta) & \sin^2(\theta) \end{bmatrix}$
- Reflection matrix:  $\begin{bmatrix} \cos(2\theta) & \sin(2\theta) \\ \sin(2\theta) & -\cos(2\theta) \end{bmatrix}$

**Gram-Schmidt process:** If  $\{x_1, x_2, \dots, x_p\}$  is a basis for a subspace  $W$  of vector space  $R^n$ , then corresponding orthogonal basis  $\{v_1, v_2, \dots, v_p\}$  for  $W$ , where

$$v_1 = x_1 \text{ and } v_i = x_i - \left(\frac{x_i \cdot v_1}{v_1 \cdot v_1}\right)v_1 - \left(\frac{x_i \cdot v_2}{v_2 \cdot v_2}\right)v_2 - \dots - \left(\frac{x_i \cdot v_{i-1}}{v_{i-1} \cdot v_{i-1}}\right)v_{i-1} \text{ for } i = 2, 3, 4, \dots, p.$$

## STATISTICS

**Moments for ungrouped data:**

- The  $r^{th}$  moment about origin:  $\mu'_r = \frac{1}{n} \sum_{i=1}^n x_i^r$ , where  $r = 1, 2, 3, \dots$ ,  $x_1, x_2, \dots, x_n$  are  $n$  observations
- The  $r^{th}$  central moment:  $\mu_r = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^r$ , where  $r = 1, 2, 3, \dots$  and  $\bar{x}$  is mean

**Moments for grouped data:**

- The  $r^{th}$  moment about origin:  $\mu'_r = \frac{1}{N} \sum_{i=1}^n f_i x_i^r$ ,  $r = 1, 2, 3, \dots$ , where observations  $x_1, x_2, \dots, x_n$  are the mid points of the class-intervals and  $f_1, f_2, \dots, f_n$  are their corresponding frequencies and  $N = \sum_{i=1}^n f_i$
- The  $r^{th}$  central moment:  $\mu_r = \frac{1}{N} \sum_{i=1}^n f_i (x_i - \bar{x})^r$ ,  $r = 1, 2, 3 \dots$  &  $\bar{x}$  is mean
- The  $r^{th}$  moment about any point A:  $\mu'_r = \frac{1}{N} \sum_{i=1}^n f_i (x_i - A)^r$ ,  $r = 1, 2, 3 \dots$

**Relation between raw moments (about origin or any point) and central moments:**

- $\mu_r = \mu'_r - {}^r C_1 \mu'_{r-1} \mu'_1 + {}^r C_2 \mu'_{r-2} \mu'^2_1 + \dots + (-1)^r \mu'^r_1$ ,  $r = 1, 2, 3 \dots$
- $\mu'_r = \mu_r + {}^r C_1 \mu_{r-1} \mu'_1 + {}^r C_2 \mu_{r-2} \mu'^2_1 + \dots + \mu'^r_1$

**Measures of kurtosis:**  $\beta_2 = \frac{\mu_4}{\mu_2^2}$

**Measures of Skewness (Karl Pearson's coefficient):**  $S_k = \frac{\sqrt{\beta_1} (\beta_2 + 3)}{2(5\beta_2 - 6\beta_1 - 9)}$ , where  $\beta_1 = \frac{\mu_3^2}{\mu_2^3}$

**Fitting of a straight line  $y = a + bx$ :** The normal equations for estimating the values of  $a$  and  $b$  are

$$\sum y = na + b \sum x, \quad \sum xy = a \sum x + b \sum x^2$$

**Fitting of a second-degree equation (quadratic)  $y = a + bx + cx^2$ :** The normal equations for estimating the values of  $a, b, c$  are

$$\begin{aligned} \sum y &= na + b \sum x + c \sum x^2, \\ \sum xy &= a \sum x + b \sum x^2 + c \sum x^3, \\ \sum x^2 y &= a \sum x^2 + b \sum x^3 + c \sum x^4. \end{aligned}$$

**Correlation coefficient (Karl Pearson correlation coefficient):**

- $r = \frac{\sum (x - \bar{x})(y - \bar{y})}{n \sigma_x \sigma_y}$ , where  $\sigma_x^2 = \frac{\sum (x - \bar{x})^2}{n}$  variance of the  $x$  series,  $\sigma_y^2 = \frac{\sum (y - \bar{y})^2}{n}$  variance of the  $y$  series,
- $\bar{x} = \frac{\sum x}{n} \rightarrow$  Mean of the  $x$  series  $\bar{y} = \frac{\sum (y - \bar{x})^2}{n} \rightarrow$  mean of the  $y$  series.
- $r = \frac{n \sum xy - (\sum x)(\sum y)}{\sqrt{\{n \sum x^2 - (\sum x)^2\} \{n \sum y^2 - (\sum y)^2\}}}$ .

**Rank correlation coefficient  $r_s$  (Spearman's rank correlation coefficient):**

- If  $x_1, x_2, \dots, x_n$  and  $y_1, y_2, \dots, y_n$  be the ranks of  $n$  individuals in characteristics A and B respectively, then  $r_s = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)}$ , where  $d_i$  is difference between ranks assigned in characteristics A and B. and  $n$  is number of pairs of data.
- Rank correlation coefficient for tied ranks:  $r_s = 1 - \frac{6 \left[ \sum_{i=1}^n d_i^2 + \frac{1}{12} (m_1^3 - m_1) + \frac{1}{12} (m_2^3 - m_2) + \dots \right]}{n(n^2 - 1)}$ , where  $m_1, m_2, \dots$  are number of repetitions of ranks.

**Linear regression:**

- Regression line of  $y$  on  $x$ :  $y - \bar{y} = b_{yx}(x - \bar{x})$ , where 
$$b_{yx} = r \frac{\sigma_y}{\sigma_x} = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2} = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$
- Regression line of  $x$  on  $y$ :  $x - \bar{x} = b_{xy}(y - \bar{y})$ , where 
$$b_{xy} = r \frac{\sigma_x}{\sigma_y} = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (y - \bar{y})^2} = \frac{n \sum xy - \sum x \sum y}{n \sum y^2 - (\sum y)^2}$$

- Angle between two lines of regression:  $\tan \theta = \frac{1-r^2}{r} \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2}$ .

**Multiple linear regression:** Fitting of a multiple linear regression model:  $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k$  for  $n$  sets of data  $(y_i, x_{ij})$ , where  $i = 1, \dots, n$  and  $j = 1, \dots, k$ . The normal equations for estimating the values of  $\beta_0, \beta_1, \beta_2, \dots, \beta_k$  are

$$\begin{aligned} n\beta_0 + \beta_1 \sum_{i=1}^n x_{i1} + \beta_2 \sum_{i=1}^n x_{i2} + \dots + \beta_k \sum_{i=1}^n x_{ik} &= \sum_{i=1}^n y_i, \\ \beta_0 \sum_{i=1}^n x_{i1} + \beta_1 \sum_{i=1}^n x_{i1}^2 + \beta_2 \sum_{i=1}^n x_{i1} x_{i2} + \dots + \beta_k \sum_{i=1}^n x_{i1} x_{ik} &= \sum_{i=1}^n x_{i1} y_i, \\ &\vdots \\ \beta_0 \sum_{i=1}^n x_{ik} + \beta_1 \sum_{i=1}^n x_{ik} x_{i1} + \beta_2 \sum_{i=1}^n x_{ik} x_{i2} + \dots + \beta_k \sum_{i=1}^n x_{ik}^2 &= \sum_{i=1}^n x_{ik} y_i. \end{aligned}$$

**Multivariate data (Sample):**

| Dimensions    | Sample                                                                                                                                                                                                                                                                                                                                                                          | Sample Mean                                                                                                                                                                                                                                                                                                                   | Sample variance (unbiased) / Sample Covariance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-dimensional | $x_1 \dots x_n$                                                                                                                                                                                                                                                                                                                                                                 | $\bar{x} = \sum_{i=1}^n x_i$                                                                                                                                                                                                                                                                                                  | $s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| p-dimensional | <p>p-random variable vector</p> $X = \begin{bmatrix} X_1 \\ \vdots \\ X_p \end{bmatrix}$ <p>where <math>x_1, x_2, \dots, x_n \in R^p</math></p> <p>n-dimensional data matrix</p> $X = \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1p} \\ x_{21} & x_{22} & \dots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \dots & x_{np} \end{bmatrix}_{n \times p}$ | <p>Sample Mean:</p> $\bar{x}_j = \frac{1}{n} \sum_{i=1}^n x_{ij}, j = 1, 2, \dots, p.$ <p>and</p> <p>Sample Mean Vector:</p> $\bar{\mathbf{x}} = \begin{bmatrix} \frac{1}{n} \sum_{i=1}^n x_{i1} \\ \vdots \\ \frac{1}{n} \sum_{i=1}^n x_{ip} \end{bmatrix} = \begin{bmatrix} \bar{x}_1 \\ \vdots \\ \bar{x}_p \end{bmatrix}$ | <p>Sample Variance:</p> $s_{jj} = s_j^2 = \frac{1}{n-1} \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2,$ <p><math>j = 1, 2, \dots, p</math> and <math>i = j</math>.</p> <p>and</p> <p>Sample Covariance:</p> $s_{jk} = s_{kj} = \frac{1}{n-1} \sum_{i=1}^n (x_{ij} - \bar{x}_j)(x_{ik} - \bar{x}_k),$ <p><math>1 \leq k, j \leq p</math> and <math>j \neq k</math>.</p> <p>Sample Covariance Matrix:</p> $S = \begin{bmatrix} s_{11} & s_{12} & \dots & s_{1p} \\ s_{21} & s_{22} & \dots & s_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ s_{p1} & s_{p2} & \dots & s_{pp} \end{bmatrix}_{p \times p}$ $= \frac{1}{n-1} \sum_{i=1}^n \begin{bmatrix} (x_{i1} - \bar{x}_1)^2 & (x_{i1} - \bar{x}_1)(x_{ip} - \bar{x}_p) \\ \vdots & \vdots \\ (x_{ip} - \bar{x}_p)(x_{i1} - \bar{x}_1) & (x_{ip} - \bar{x}_p)^2 \end{bmatrix}_{p \times p}$ |

**Multivariate data (Population):**

| Dimensions    | Population                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Population Mean                                                                                                                                                                                                                                                                                                                  | Population variance / Population Covariance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-dimensional | $x_1 \dots x_n$                                                                                                                                                                                                                                                                                                                                                                                                                                                        | $\mu = \sum_{i=1}^n x_i$                                                                                                                                                                                                                                                                                                         | $\sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| p-dimensional | <p>p-random variable vector</p> $X = \begin{bmatrix} X_1 \\ \vdots \\ X_p \end{bmatrix}$ <p>where <math>x_1 \dots x_n \in R^p</math></p> <p>n-dimensional data matrix</p> $X = \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1p} \\ x_{21} & x_{22} & \dots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \dots & x_{np} \end{bmatrix}_{n \times p}$ <p>Here each column in the data matrix corresponds to a random variable <math>X_j</math>.</p> | <p>Population Mean:</p> $\mu_j = \frac{1}{n} \sum_{i=1}^n x_{ij},$ <p><math>j = 1, 2, \dots, p.</math></p> <p>and</p> <p>Population Mean Vector:</p> $\mu = \begin{bmatrix} \frac{1}{n} \sum_{i=1}^n x_{i1} \\ \vdots \\ \frac{1}{n} \sum_{i=1}^n x_{ip} \end{bmatrix} = \begin{bmatrix} \mu_1 \\ \vdots \\ \mu_p \end{bmatrix}$ | <p>Population Variance:</p> $\sigma_{jj} = \sigma_j^2 = \frac{1}{n} \sum_{i=1}^n (x_{ij} - \mu_j)^2,$ <p><math>j = 1, 2, \dots, p</math> and <math>i = j.</math></p> <p>and</p> <p>Population Covariance:</p> $\sigma_{jk} = \sigma_{kj} = \frac{1}{n} \sum_{i=1}^n (x_{ij} - \mu_j)(x_{ik} - \mu_k),$ <p><math>1 \leq k, j \leq p</math> and <math>j \neq k.</math></p> <p>Population Covariance Matrix:</p> $\Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \dots & \sigma_{1p} \\ \sigma_{21} & \sigma_{22} & \dots & \sigma_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{p1} & \sigma_{p2} & \dots & \sigma_{pp} \end{bmatrix}_{p \times p}$ $= \frac{1}{n} \sum_{i=1}^n \begin{bmatrix} (x_{i1} - \mu_1) & \dots & (x_{i1} - \mu_1) & (x_{ip} - \mu_p) \\ \vdots & & \vdots & \vdots \\ (x_{ip} - \mu_p) & (x_{i1} - \mu_1) & \dots & (x_{ip} - \mu_p) \end{bmatrix}_{p \times p}$ |

**Rank Correlation, Partial Correlation, Multiple Correlation**

In the case of tri variate data  $X_1, X_2$  and  $X_3$

i) Coefficient of partial correlation of  $X_1$  and  $X_2$ , keeping  $X_3$  constant:

$$r_{12.3} = \frac{r_{12} - r_{13}r_{23}}{\sqrt{1 - r_{13}^2}\sqrt{1 - r_{23}^2}}$$

ii) Coefficient of partial correlation of  $X_1$  and  $X_3$ , keeping  $X_2$  constant:

$$r_{13.2} = \frac{r_{13} - r_{12}r_{23}}{\sqrt{1 - r_{12}^2}\sqrt{1 - r_{23}^2}}$$

iii) Coefficient of partial correlation of  $X_2$  and  $X_3$ , keeping  $X_1$  constant:

$$r_{23.1} = \frac{r_{23} - r_{12}r_{13}}{\sqrt{1 - r_{12}^2}\sqrt{1 - r_{13}^2}}$$

**Note:** In all the above three cases  $r_{ij} = r_{ji}$  for all  $i \neq j$ .

In the case of tri variate data:  $X_1, X_2$  and  $X_3$

- Multiple correlation coefficient of  $X_1$  on  $X_2$  and  $X_3$ :  $R_{1.23} = \sqrt{\frac{r_{12}^2 + r_{13}^2 - 2r_{12}r_{13}r_{23}}{1 - r_{23}^2}}$
- Multiple correlation coefficient of  $X_2$  on  $X_1$  and  $X_3$ :  $R_{2.13} = \sqrt{\frac{r_{21}^2 + r_{23}^2 - 2r_{21}r_{23}r_{13}}{1 - r_{13}^2}}$
- Multiple correlation coefficient of  $X_3$  on  $X_1$  and  $X_2$ :  $R_{3.12} = \sqrt{\frac{r_{31}^2 + r_{32}^2 - 2r_{31}r_{32}r_{12}}{1 - r_{12}^2}}$

**Given variables  $x, y$  and  $z$ , we define the multiple correlation coefficient.**

$$R_{z.xy} = \sqrt{\frac{r_{zx}^2 + r_{zy}^2 - 2r_{zx}r_{zy}r_{yx}}{1 - r_{yx}^2}} \text{ and } r_{xy} = \frac{n\sum xy - \sum x \sum y}{\sqrt{\{n(\sum x^2) - (\sum x)^2\}\{n(\sum y^2) - (\sum y)^2\}}}$$

**Note:** In all the above three cases  $r_{ij} = r_{ji}$  for all  $i \neq j$ .

## Analysis of Variance (ANOVA)

### One-way ANOVA Table

| Sources of variation | Degrees of freedom | Sum of squares | Mean sum of squares | $F_{cal}$                     |
|----------------------|--------------------|----------------|---------------------|-------------------------------|
| Between classes      | $k - 1$            | SSB            | MSSB                | $F_{cal} = \frac{MSSB}{MSSW}$ |
| Within classes       | $N - k$            | SSW            | MSSW                |                               |
| Total                | $N - 1$            | SST            |                     |                               |

$k$ : number of classes,  $N = pq$  is number of rows  $\times$  number of columns,  $n$ : number of entries in each  $X_i$

$T$ : Grand sum:  $\sum X_i$ , Correction factor (C.F.) =  $\frac{T^2}{N}$

SSB: Sum of squares between classes =  $\frac{(\sum X_i)^2}{n_i} - C.F.$

SST: Total sum of squares =  $\sum X_i^2 - C.F.$

SSW: Sum of squares within classes =  $SST - SSB$

MSSB: Mean sum of squares between classes =  $\frac{SSB}{k-1}$

MSSW: Mean sum of squares within classes =  $\frac{SSW}{N-k}$

**Conclusion:**

If  $F_{cal} < F_{tab}$ , there is NO significant difference, Null Hypothesis  $H_0$  is accepted.

If  $F_{cal} > F_{tab}$ , there is a significant difference, Null Hypothesis  $H_0$  is rejected and Alternative Hypothesis,  $H_1$  is accepted.

Note: In one-way ANOVA,  $F_{cal}$  can also be computed using columns.

Two-way ANOVA table

| Source of variation | Degrees of freedom | Sum of squares | Mean sum of squares | $F_{cal}$                               |
|---------------------|--------------------|----------------|---------------------|-----------------------------------------|
| Between rows        | $r - 1$            | SSR            | MSSR                | $F_{cal (rows)} = \frac{MSSR}{MSSE}$    |
| Between columns     | $c - 1$            | SSC            | MSSC                | $F_{cal (columns)} = \frac{MSSC}{MSSE}$ |
| Error (residual)    | $(r - 1)(c - 1)$   | SSE            | MSSE                |                                         |
| Total               | $rc - 1$           | SST            |                     |                                         |

Number of levels of row factor =  $r$ , Number of levels of column factor =  $c$

Total number of observations =  $rc$  = (rows  $\times$  columns)

Observations in  $ij^{th}$  cell of table =  $x_{ij}$ , Sum of  $c$  observations in  $i^{th}$  row =  $T_{Ri} = \sum_j x_{ij}$

Sum of  $r$  observations in  $j^{th}$  column =  $T_{Cj} = \sum_i x_{ij}$

Sum of all observations =  $T = \sum_i T_{Ri} = \sum_j T_{Cj}$

Correction factor =  $C.F. = \frac{T^2}{rc}$

Sum of squares between rows:  $SSR = \sum \frac{T_{Ri}^2}{c} - C.F.$

Sum of squares between columns:  $SSC = \sum \frac{T_{Cj}^2}{r} - C.F.$

Total sum of squares:  $SST = \sum_i \sum_j x_{ij}^2 - C.F.$

Error (residual) sum of squares:  $SSE = SST - SSR - SSC$

MSSR = mean sum of squares between rows =  $\frac{SSR}{r-1}$

MSSC = mean sum of squares between columns =  $\frac{SSC}{c-1}$

$$MSSE = \text{mean sum of squares of error} = \frac{SSE}{(r-1)(c-1)}$$

### Conclusion:

If  $F_{cal} < F_{tab}$ , there is no significant difference, null hypothesis  $H_0$  is accepted.

If  $F_{cal} > F_{tab}$ , there is a significant difference, null hypothesis  $H_0$  is rejected and alternative hypothesis,  $H_1$  is accepted.

## PROBABILITY AND RANDOM VARIABLES

**Probability of an event:** If a trial results in  $n$  exhaustive, mutually exclusive and equally likely cases and  $m$  of them are favourable to the happening of an event  $E$  then the probability 'p' of happening of  $E$  is given by  $p = P(E) = \frac{m}{n}$

### Addition theorem:

**For two events:** If  $A$  and  $B$  are any two events with respective probabilities  $P(A)$  and  $P(B)$ , then the probability of occurrence of at least one of the events is given by

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

**For three events:** If  $A, B$  and  $C$  are any three events with respective probabilities  $P(A), P(B)$  and  $P(C)$ , then the probability of occurrence of at least one of the events is given by

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$$

**Conditional probability:** Conditional probability of  $B$ , given  $A$ , denoted by  $P(B|A)$ , is defined by  $P(A|B) = \frac{P(A \cap B)}{P(A)}$ , provided  $P(A) > 0$ .

**Multiplication rule:** Suppose  $A$  and  $B$  are events in a sample space  $S$  with  $P(A) > 0$  multiplication rule is:  $P(A \cap B) = P(A)P(B|A)$

**Bayes' theorem (rule):** If  $B_1, B_2, \dots, B_n$  are mutually disjoint events with  $P(B_i) \neq 0$  ( $i = 1, 2, \dots, n$ ) then for any arbitrary event  $A$  which is a subset of  $\bigcup_{i=1}^n B_i$  such that  $P(A) > 0$ , then  $P(B_i|A) = \frac{P(B_i)P(A|B_i)}{\sum_{i=1}^n P(B_i)P(A|B_i)}$ .

**Discrete random variable:** Let  $X$  be a discrete random variable. A function  $p(x)$  is a probability mass function of the discrete random variable  $X$  if  $p(x) \geq 0, \forall x \in X$  and  $\sum_x p(x) = 1$ .

- Expectation,  $E(X) = \sum_x xp(x)$
- If  $Y = g(X)$ , then  $E(Y) = \sum_x g(x)p(x)$
- Variance,  $Var(X) = E[(X - E(X))^2] = E(X^2) - [E(X)]^2$
- Standard deviation,  $\sigma_X = \sqrt{Var(X)}$
- The cumulative distribution function,  $F(t) = P(X \leq t) = \sum_{x \leq t} p(x)$

**Continuous random variable:** Suppose  $X$  is a continuous random variable. A function  $f(x)$  is called a probability density function of the continuous random variable  $X$  if  $f(x) \geq 0 \forall x \in X$  and  $\int_{-\infty}^{\infty} f(x)dx = 1$ .

- Expectation,  $E(X) = \int_{-\infty}^{\infty} x f(x)dx$
- If  $Y = g(X)$ , then  $E(Y) = \int_{-\infty}^{\infty} g(x) f(x)dx$
- Variance,  $Var(X) = E[(X - E(X))^2] = E(X^2) - [E(X)]^2$
- Standard deviation,  $\sigma_X = \sqrt{Var(X)}$
- Cumulative distribution function,  $F(t) = P(X \leq t) = \int_{-\infty}^t f(x)dx$
- $P(a \leq X \leq b) = P(a \leq X < b) = P(a < X \leq b) = P(a < X < b) = \int_a^b f(x)dx$

**Markov's inequality:** For any random variable  $X$  with finite  $E[X]$  and any  $k > 0$ , the probability that  $X$  is at least  $k$  times its expected value is at-most  $\frac{1}{k}$ . That is,  $P[X \geq kE[X]] \leq \frac{1}{k}$  or  $P[X \geq k] \leq \frac{E[X]}{k}$ .

**Chebyshev's inequality:** Any random variable  $X$  with expectation  $\mu = E[X]$  and variance  $\sigma^2 = Var[X]$  belongs to the interval  $\mu \pm k = [\mu - k, \mu + k]$  with probability of at least  $1 - \left(\frac{\sigma}{k}\right)^2$ . That is,  $P\{|X - \mu| \geq k\} \leq \left(\frac{\sigma}{k}\right)^2 = \frac{\sigma^2}{k^2}$

**Joint probability mass function:** Suppose  $X$  and  $Y$  are two discrete random variables. A function  $p(x, y)$  is called a joint probability mass function of  $X$  and  $Y$  if  $p(x, y) \geq 0, \forall x \in X, y \in Y$  and  $\sum_x \sum_y p(x, y) = 1$ .

- Let  $Z = g(X, Y)$ . Expectation,  $E[Z] = \sum_x \sum_y g(x, y)p(x, y)$
- The marginal distributions of  $X$  alone and of  $Y$  alone are:  $g(x) = \sum_y p(x, y)$  and  $h(y) = \sum_x p(x, y)$
- Covariance,  $Cov(X, Y) = E(XY) - E(X)E(Y)$
- Correlation of  $X$  and  $Y$ ,  $\rho(X, Y) = \frac{Cov(X, Y)}{\sigma_X \sigma_Y}$
- For discrete random variables  $X$  and  $Y$  with joint pmf  $p(x, y)$  and  $x, y$  such that  $g(x) > 0, h(y) > 0$ , then the conditional probability mass functions are  
(i)  $P(X = x|Y = y) = \frac{p(x, y)}{h(y)}$ , (ii)  $P(Y = y|X = x) = \frac{p(x, y)}{g(x)}$ .
- If  $X$  and  $Y$  are independent, then  $E(XY) = E(X)E(Y)$

**Joint probability density function:** Suppose  $X$  and  $Y$  are two continuous random variables. A function  $f(x, y)$  is called a joint probability density function of  $X$  and  $Y$  if  $f(x, y) \geq 0, \forall x \in X, y \in Y$  and  $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$ .

- Let  $Z = g(X, Y)$ . Expectation,  $E[Z] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y)f(x, y) dx dy$
- The marginal distributions of  $X$  alone and of  $Y$  alone are  $g(x) = \int_{-\infty}^{\infty} f(x, y) dy$  and  $h(y) = \int_{-\infty}^{\infty} f(x, y) dx$
- Covariance,  $Cov(X, Y) = E(XY) - E(X)E(Y)$
- Correlation of  $X$  and  $Y$ ,  $\rho(X, Y) = \frac{Cov(X, Y)}{\sigma_X \sigma_Y}$
- If  $X$  and  $Y$  are independent, then  $E(XY) = E(X)E(Y)$

- The conditional distribution of the random variable  $Y$  given that  $X = x$  is  $f(y|x) = \frac{f(x,y)}{g(x)}$ , provided  $g(x) > 0$
- The conditional distribution of the random variable  $X$  given that  $Y = y$  is  $f(x|y) = \frac{f(x,y)}{h(y)}$ , provided  $h(y) > 0$ .

## PROBABILITY DISTRIBUTIONS

### **Bernoulli distribution:**

- The probability mass function is given by  $f(x, p) = p^x(1 - p)^{1-x}$ ;  $x \in (0,1)$
- Mean,  $\mu = p$
- Variance,  $\sigma^2 = pq$
- Standard deviation,  $\sigma = \sqrt{pq}$

### **Binomial distribution:**

- The probability function of the binomial distribution is given by  $b(x; n, p) = nC_x p^x q^{n-x}$ , where  $x = 1, 2, 3, \dots$ ,  $p$  is the probability of success and  $q = 1 - p$  is the probability of failure.
- Mean,  $\mu = np$
- Variance,  $\sigma^2 = npq$
- Standard deviation,  $\sigma = \sqrt{npq}$

### **Poisson distribution:**

- The probability function of the Poisson distribution is given by  $p(x; \lambda) = P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}$ , where  $\lambda$  is the parameter of the Poisson distribution.
- Mean,  $\mu = \lambda$
- Variance,  $\sigma^2 = \lambda$
- Standard deviation,  $\sigma = \sqrt{\lambda}$

### **Geometric distribution:**

- Probability mass function is given by  $P(X = x) = (1 - p)^{x-1}p$ , where  $0 < p < 1$
- Mean,  $\mu = 1/p$
- Variance,  $\sigma^2 = \frac{1-p}{p^2}$
- Standard deviation  $\sigma = \frac{\sqrt{1-p}}{p}$

**Exponential distribution:** A continuous random variable  $X$  assuming non-negative values is said to have an exponential distribution with parameter  $\lambda > 0$ , if its probability density function is given by

$$f(x; \lambda) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

- Mean  $\mu = \frac{1}{\lambda}$

- Variance  $\sigma^2 = \left(\frac{1}{\lambda}\right)^2$
- Standard deviation  $\sigma = \frac{1}{\lambda}$

**Uniform distribution:** The probability density function is  $f(x) = \frac{1}{b-a}$  for  $a \leq x \leq b$

- Mean  $\mu = \frac{a+b}{2}$
- Variance  $\sigma^2 = \frac{(b-a)^2}{12}$
- Standard deviation  $\sigma = \sqrt{\frac{(b-a)^2}{12}}$

**Normal distribution:** A random variable  $X$  is said to have a normal distribution with parameters  $\mu$  (called "mean") and  $\sigma^2$  (called "variance") if its density function is given by the probability law:

$$f(x; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2} \left(\frac{x-\mu}{\sigma}\right)^2\right] \text{ for } -\infty < x < \infty, -\infty < \mu < \infty \text{ and } 0 < \sigma < \infty.$$

- Standard normal distribution or  $z$  – distribution is given by

$$f(z; 0, 1) = \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{z^2}{2}\right], \quad -\infty < z < \infty,$$

- Cumulative standard normal distribution is

$$\Phi(z) = P(Z \leq z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}u^2} du.$$

Note: The value of the integral can be calculated by the Normal distribution table.

## **SAMPLING THEORY**

**Numerical summation of data:** If the  $n$  observations in a sample are denoted by  $x_1, x_2, \dots, x_n$ , then

- Sample mean:  $\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$
- Sample variance:  $s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$
- Standard deviation is the positive square root of the sample variance.
- If the population is finite with size  $N$ , then for the sampling distribution of  $\bar{x}$ :

Mean:  $\mu_{\bar{x}} = \mu$

Variance:  $\sigma_{\bar{x}}^2 = \frac{\sigma^2}{n} \times \frac{N-n}{N-1}$ , if the sampling is without replacement

$\sigma_{\bar{x}}^2 = \frac{\sigma^2}{n}$ , if the sampling is with replacement or population is infinite.

**Sampling distributions:** Let  $X$  be a random variable of a population with mean  $\mu$  and variance  $\sigma^2$ . If the random variables  $X_1, X_2, \dots, X_n$  are a random sample on size  $n$ , then sample mean  $\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n}$

**Sampling distribution of means**

- Mean:  $\mu_{\bar{X}} = \mu$
- Variance:  $\sigma_{\bar{X}}^2 = \frac{\sigma^2}{n}$

- If the population is normal or sample size is sufficiently large, then the distribution of  $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$  is approximately standard normal.

### Sampling distribution of a difference in sample means

- Mean:  $\mu_{(\bar{X}_1 - \bar{X}_2)} = \mu_{\bar{X}_1} - \mu_{\bar{X}_2} = \mu_1 - \mu_2$
- Variance:  $\sigma_{(\bar{X}_1 - \bar{X}_2)}^2 = \sigma_{\bar{X}_1}^2 + \sigma_{\bar{X}_2}^2 = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$
- If the populations are normal or sample sizes are sufficiently large, then the distribution of  $Z = \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$  is approximately standard normal.

### Sampling distribution of proportions:

- Mean  $\mu_{\hat{p}} = p$
- Standard deviation  $\sigma_{\hat{p}} = \sqrt{\frac{pq}{n}} = \left[ \frac{p(1-p)}{n} \right]^{\frac{1}{2}}$
- The distribution of  $Z = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}}$  is approximately standard normal if  $n$  is large or  $np \geq 5$  and  $np(1-p) \geq 5$ .

### Statistical decisions:

- Levels of significance for two – tailed test

| Level of significance | Critical value | Acceptance region |
|-----------------------|----------------|-------------------|
| 0.05                  | $z_c = 1.96$   | $(-1.96, 1.96)$   |
| 0.01                  | $z_c = 2.58$   | $(-2.58, 2.58)$   |

- Levels of significance for one– tailed test

| Level of significance | Critical value    |                  | Critical region   |                     |
|-----------------------|-------------------|------------------|-------------------|---------------------|
|                       | Right-tailed test | Left-tailed test | Right-tailed test | Left-tailed test    |
| 0.05                  | $z_c = 1.645$     | $-z_c = -1.645$  | $(1.645, \infty)$ | $(-\infty, -1.645)$ |
| 0.01                  | $z_c = 2.33$      | $-z_c = -2.33$   | $(2.33, \infty)$  | $(-\infty, -2.33)$  |

- For the standard normal variate  $Z$ , and the test value of  $Z$  is  $z_0$ , then the P-value is determined as follows:
  - For left-tail test, the P-value is  $P(Z \leq z_0)$
  - For right-tail test, the P-value is  $P(Z \geq z_0)$
  - For two tail test, the P-value is  $P(Z \leq -|z_0|) + P(Z \geq |z_0|) = 2(1 - \phi(|z_0|))$

### Test of significance:

#### z- test:

- Test statistics:
  - $z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$ , if  $\sigma$  is known.

(ii)  $z = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$ , if  $\sigma$  is unknown and  $n$  is large.

**t-test:**

- Test statistic:

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}, \text{ if } \sigma \text{ is unknown.}$$

**Chi – square ( $\chi^2$ ) test:**

- If  $s^2$  is the variance of a random sample of size  $n$  taken from a normal population having variance  $\sigma^2$ , then the statistic

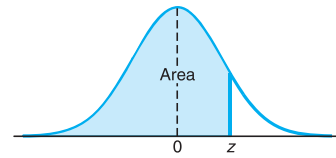
$$\chi^2 = \frac{(n-1)s^2}{\sigma^2} = \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{\sigma^2}$$

is a chi-squared distribution with  $\nu = n - 1$  degrees of freedom.

- If  $f_1, f_2, f_3, \dots, f_n$  are the observed frequencies and  $e_1, e_2, e_3, \dots, e_n$  are the expected or theoretical frequencies. The statistic  $\chi^2 = \sum_{i=1}^n \frac{(f_i - e_i)^2}{e_i}$

**F- test:**

- Test statistic:  $F_0 = \frac{S_1^2}{S_2^2}$

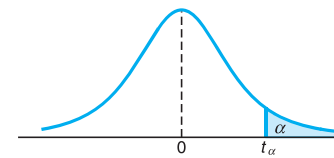


Areas under the Normal Curve

| $z$  | .00    | .01    | .02    | .03    | .04    | .05    | .06    | .07    | .08    | .09    |
|------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| −3.4 | 0.0003 | 0.0003 | 0.0003 | 0.0003 | 0.0003 | 0.0003 | 0.0003 | 0.0003 | 0.0003 | 0.0002 |
| −3.3 | 0.0005 | 0.0005 | 0.0005 | 0.0004 | 0.0004 | 0.0004 | 0.0004 | 0.0004 | 0.0004 | 0.0003 |
| −3.2 | 0.0007 | 0.0007 | 0.0006 | 0.0006 | 0.0006 | 0.0006 | 0.0006 | 0.0005 | 0.0005 | 0.0005 |
| −3.1 | 0.0010 | 0.0009 | 0.0009 | 0.0009 | 0.0008 | 0.0008 | 0.0008 | 0.0008 | 0.0007 | 0.0007 |
| −3.0 | 0.0013 | 0.0013 | 0.0013 | 0.0012 | 0.0012 | 0.0011 | 0.0011 | 0.0011 | 0.0010 | 0.0010 |
| −2.9 | 0.0019 | 0.0018 | 0.0018 | 0.0017 | 0.0016 | 0.0016 | 0.0015 | 0.0015 | 0.0014 | 0.0014 |
| −2.8 | 0.0026 | 0.0025 | 0.0024 | 0.0023 | 0.0023 | 0.0022 | 0.0021 | 0.0021 | 0.0020 | 0.0019 |
| −2.7 | 0.0035 | 0.0034 | 0.0033 | 0.0032 | 0.0031 | 0.0030 | 0.0029 | 0.0028 | 0.0027 | 0.0026 |
| −2.6 | 0.0047 | 0.0045 | 0.0044 | 0.0043 | 0.0041 | 0.0040 | 0.0039 | 0.0038 | 0.0037 | 0.0036 |
| −2.5 | 0.0062 | 0.0060 | 0.0059 | 0.0057 | 0.0055 | 0.0054 | 0.0052 | 0.0051 | 0.0049 | 0.0048 |
| −2.4 | 0.0082 | 0.0080 | 0.0078 | 0.0075 | 0.0073 | 0.0071 | 0.0069 | 0.0068 | 0.0066 | 0.0064 |
| −2.3 | 0.0107 | 0.0104 | 0.0102 | 0.0099 | 0.0096 | 0.0094 | 0.0091 | 0.0089 | 0.0087 | 0.0084 |
| −2.2 | 0.0139 | 0.0136 | 0.0132 | 0.0129 | 0.0125 | 0.0122 | 0.0119 | 0.0116 | 0.0113 | 0.0110 |
| −2.1 | 0.0179 | 0.0174 | 0.0170 | 0.0166 | 0.0162 | 0.0158 | 0.0154 | 0.0150 | 0.0146 | 0.0143 |
| −2.0 | 0.0228 | 0.0222 | 0.0217 | 0.0212 | 0.0207 | 0.0202 | 0.0197 | 0.0192 | 0.0188 | 0.0183 |
| −1.9 | 0.0287 | 0.0281 | 0.0274 | 0.0268 | 0.0262 | 0.0256 | 0.0250 | 0.0244 | 0.0239 | 0.0233 |
| −1.8 | 0.0359 | 0.0351 | 0.0344 | 0.0336 | 0.0329 | 0.0322 | 0.0314 | 0.0307 | 0.0301 | 0.0294 |
| −1.7 | 0.0446 | 0.0436 | 0.0427 | 0.0418 | 0.0409 | 0.0401 | 0.0392 | 0.0384 | 0.0375 | 0.0367 |
| −1.6 | 0.0548 | 0.0537 | 0.0526 | 0.0516 | 0.0505 | 0.0495 | 0.0485 | 0.0475 | 0.0465 | 0.0455 |
| −1.5 | 0.0668 | 0.0655 | 0.0643 | 0.0630 | 0.0618 | 0.0606 | 0.0594 | 0.0582 | 0.0571 | 0.0559 |
| −1.4 | 0.0808 | 0.0793 | 0.0778 | 0.0764 | 0.0749 | 0.0735 | 0.0721 | 0.0708 | 0.0694 | 0.0681 |
| −1.3 | 0.0968 | 0.0951 | 0.0934 | 0.0918 | 0.0901 | 0.0885 | 0.0869 | 0.0853 | 0.0838 | 0.0823 |
| −1.2 | 0.1151 | 0.1131 | 0.1112 | 0.1093 | 0.1075 | 0.1056 | 0.1038 | 0.1020 | 0.1003 | 0.0985 |
| −1.1 | 0.1357 | 0.1335 | 0.1314 | 0.1292 | 0.1271 | 0.1251 | 0.1230 | 0.1210 | 0.1190 | 0.1170 |
| −1.0 | 0.1587 | 0.1562 | 0.1539 | 0.1515 | 0.1492 | 0.1469 | 0.1446 | 0.1423 | 0.1401 | 0.1379 |
| −0.9 | 0.1841 | 0.1814 | 0.1788 | 0.1762 | 0.1736 | 0.1711 | 0.1685 | 0.1660 | 0.1635 | 0.1611 |
| −0.8 | 0.2119 | 0.2090 | 0.2061 | 0.2033 | 0.2005 | 0.1977 | 0.1949 | 0.1922 | 0.1894 | 0.1867 |
| −0.7 | 0.2420 | 0.2389 | 0.2358 | 0.2327 | 0.2296 | 0.2266 | 0.2236 | 0.2206 | 0.2177 | 0.2148 |
| −0.6 | 0.2743 | 0.2709 | 0.2676 | 0.2643 | 0.2611 | 0.2578 | 0.2546 | 0.2514 | 0.2483 | 0.2451 |
| −0.5 | 0.3085 | 0.3050 | 0.3015 | 0.2981 | 0.2946 | 0.2912 | 0.2877 | 0.2843 | 0.2810 | 0.2776 |
| −0.4 | 0.3446 | 0.3409 | 0.3372 | 0.3336 | 0.3300 | 0.3264 | 0.3228 | 0.3192 | 0.3156 | 0.3121 |
| −0.3 | 0.3821 | 0.3783 | 0.3745 | 0.3707 | 0.3669 | 0.3632 | 0.3594 | 0.3557 | 0.3520 | 0.3483 |
| −0.2 | 0.4207 | 0.4168 | 0.4129 | 0.4090 | 0.4052 | 0.4013 | 0.3974 | 0.3936 | 0.3897 | 0.3859 |
| −0.1 | 0.4602 | 0.4562 | 0.4522 | 0.4483 | 0.4443 | 0.4404 | 0.4364 | 0.4325 | 0.4286 | 0.4247 |
| −0.0 | 0.5000 | 0.4960 | 0.4920 | 0.4880 | 0.4840 | 0.4801 | 0.4761 | 0.4721 | 0.4681 | 0.4641 |



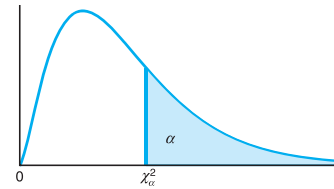
Critical Values of the  $t$ -Distribution



| $v$      | $\alpha$ |       |       |       |       |       |        |
|----------|----------|-------|-------|-------|-------|-------|--------|
|          | 0.40     | 0.30  | 0.20  | 0.15  | 0.10  | 0.05  | 0.025  |
| 1        | 0.325    | 0.727 | 1.376 | 1.963 | 3.078 | 6.314 | 12.706 |
| 2        | 0.289    | 0.617 | 1.061 | 1.386 | 1.886 | 2.920 | 4.303  |
| 3        | 0.277    | 0.584 | 0.978 | 1.250 | 1.638 | 2.353 | 3.182  |
| 4        | 0.271    | 0.569 | 0.941 | 1.190 | 1.533 | 2.132 | 2.776  |
| 5        | 0.267    | 0.559 | 0.920 | 1.156 | 1.476 | 2.015 | 2.571  |
| 6        | 0.265    | 0.553 | 0.906 | 1.134 | 1.440 | 1.943 | 2.447  |
| 7        | 0.263    | 0.549 | 0.896 | 1.119 | 1.415 | 1.895 | 2.365  |
| 8        | 0.262    | 0.546 | 0.889 | 1.108 | 1.397 | 1.860 | 2.306  |
| 9        | 0.261    | 0.543 | 0.883 | 1.100 | 1.383 | 1.833 | 2.262  |
| 10       | 0.260    | 0.542 | 0.879 | 1.093 | 1.372 | 1.812 | 2.228  |
| 11       | 0.260    | 0.540 | 0.876 | 1.088 | 1.363 | 1.796 | 2.201  |
| 12       | 0.259    | 0.539 | 0.873 | 1.083 | 1.356 | 1.782 | 2.179  |
| 13       | 0.259    | 0.538 | 0.870 | 1.079 | 1.350 | 1.771 | 2.160  |
| 14       | 0.258    | 0.537 | 0.868 | 1.076 | 1.345 | 1.761 | 2.145  |
| 15       | 0.258    | 0.536 | 0.866 | 1.074 | 1.341 | 1.753 | 2.131  |
| 16       | 0.258    | 0.535 | 0.865 | 1.071 | 1.337 | 1.746 | 2.120  |
| 17       | 0.257    | 0.534 | 0.863 | 1.069 | 1.333 | 1.740 | 2.110  |
| 18       | 0.257    | 0.534 | 0.862 | 1.067 | 1.330 | 1.734 | 2.101  |
| 19       | 0.257    | 0.533 | 0.861 | 1.066 | 1.328 | 1.729 | 2.093  |
| 20       | 0.257    | 0.533 | 0.860 | 1.064 | 1.325 | 1.725 | 2.086  |
| 21       | 0.257    | 0.532 | 0.859 | 1.063 | 1.323 | 1.721 | 2.080  |
| 22       | 0.256    | 0.532 | 0.858 | 1.061 | 1.321 | 1.717 | 2.074  |
| 23       | 0.256    | 0.532 | 0.858 | 1.060 | 1.319 | 1.714 | 2.069  |
| 24       | 0.256    | 0.531 | 0.857 | 1.059 | 1.318 | 1.711 | 2.064  |
| 25       | 0.256    | 0.531 | 0.856 | 1.058 | 1.316 | 1.708 | 2.060  |
| 26       | 0.256    | 0.531 | 0.856 | 1.058 | 1.315 | 1.706 | 2.056  |
| 27       | 0.256    | 0.531 | 0.855 | 1.057 | 1.314 | 1.703 | 2.052  |
| 28       | 0.256    | 0.530 | 0.855 | 1.056 | 1.313 | 1.701 | 2.048  |
| 29       | 0.256    | 0.530 | 0.854 | 1.055 | 1.311 | 1.699 | 2.045  |
| 30       | 0.256    | 0.530 | 0.854 | 1.055 | 1.310 | 1.697 | 2.042  |
| 40       | 0.255    | 0.529 | 0.851 | 1.050 | 1.303 | 1.684 | 2.021  |
| 60       | 0.254    | 0.527 | 0.848 | 1.045 | 1.296 | 1.671 | 2.000  |
| 120      | 0.254    | 0.526 | 0.845 | 1.041 | 1.289 | 1.658 | 1.980  |
| $\infty$ | 0.253    | 0.524 | 0.842 | 1.036 | 1.282 | 1.645 | 1.960  |

(continued) Critical Values of the  $t$ -Distribution

| $v$      | $\alpha$ |        |        |        |        |         |         |
|----------|----------|--------|--------|--------|--------|---------|---------|
|          | 0.02     | 0.015  | 0.01   | 0.0075 | 0.005  | 0.0025  | 0.0005  |
| 1        | 15.894   | 21.205 | 31.821 | 42.433 | 63.656 | 127.321 | 636.578 |
| 2        | 4.849    | 5.643  | 6.965  | 8.073  | 9.925  | 14.089  | 31.600  |
| 3        | 3.482    | 3.896  | 4.541  | 5.047  | 5.841  | 7.453   | 12.924  |
| 4        | 2.999    | 3.298  | 3.747  | 4.088  | 4.604  | 5.598   | 8.610   |
| 5        | 2.757    | 3.003  | 3.365  | 3.634  | 4.032  | 4.773   | 6.869   |
| 6        | 2.612    | 2.829  | 3.143  | 3.372  | 3.707  | 4.317   | 5.959   |
| 7        | 2.517    | 2.715  | 2.998  | 3.203  | 3.499  | 4.029   | 5.408   |
| 8        | 2.449    | 2.634  | 2.896  | 3.085  | 3.355  | 3.833   | 5.041   |
| 9        | 2.398    | 2.574  | 2.821  | 2.998  | 3.250  | 3.690   | 4.781   |
| 10       | 2.359    | 2.527  | 2.764  | 2.932  | 3.169  | 3.581   | 4.587   |
| 11       | 2.328    | 2.491  | 2.718  | 2.879  | 3.106  | 3.497   | 4.437   |
| 12       | 2.303    | 2.461  | 2.681  | 2.836  | 3.055  | 3.428   | 4.318   |
| 13       | 2.282    | 2.436  | 2.650  | 2.801  | 3.012  | 3.372   | 4.221   |
| 14       | 2.264    | 2.415  | 2.624  | 2.771  | 2.977  | 3.326   | 4.140   |
| 15       | 2.249    | 2.397  | 2.602  | 2.746  | 2.947  | 3.286   | 4.073   |
| 16       | 2.235    | 2.382  | 2.583  | 2.724  | 2.921  | 3.252   | 4.015   |
| 17       | 2.224    | 2.368  | 2.567  | 2.706  | 2.898  | 3.222   | 3.965   |
| 18       | 2.214    | 2.356  | 2.552  | 2.689  | 2.878  | 3.197   | 3.922   |
| 19       | 2.205    | 2.346  | 2.539  | 2.674  | 2.861  | 3.174   | 3.883   |
| 20       | 2.197    | 2.336  | 2.528  | 2.661  | 2.845  | 3.153   | 3.850   |
| 21       | 2.189    | 2.328  | 2.518  | 2.649  | 2.831  | 3.135   | 3.819   |
| 22       | 2.183    | 2.320  | 2.508  | 2.639  | 2.819  | 3.119   | 3.792   |
| 23       | 2.177    | 2.313  | 2.500  | 2.629  | 2.807  | 3.104   | 3.768   |
| 24       | 2.172    | 2.307  | 2.492  | 2.620  | 2.797  | 3.091   | 3.745   |
| 25       | 2.167    | 2.301  | 2.485  | 2.612  | 2.787  | 3.078   | 3.725   |
| 26       | 2.162    | 2.296  | 2.479  | 2.605  | 2.779  | 3.067   | 3.707   |
| 27       | 2.158    | 2.291  | 2.473  | 2.598  | 2.771  | 3.057   | 3.689   |
| 28       | 2.154    | 2.286  | 2.467  | 2.592  | 2.763  | 3.047   | 3.674   |
| 29       | 2.150    | 2.282  | 2.462  | 2.586  | 2.756  | 3.038   | 3.660   |
| 30       | 2.147    | 2.278  | 2.457  | 2.581  | 2.750  | 3.030   | 3.646   |
| 40       | 2.123    | 2.250  | 2.423  | 2.542  | 2.704  | 2.971   | 3.551   |
| 60       | 2.099    | 2.223  | 2.390  | 2.504  | 2.660  | 2.915   | 3.460   |
| 120      | 2.076    | 2.196  | 2.358  | 2.468  | 2.617  | 2.860   | 3.373   |
| $\infty$ | 2.054    | 2.170  | 2.326  | 2.432  | 2.576  | 2.807   | 3.290   |



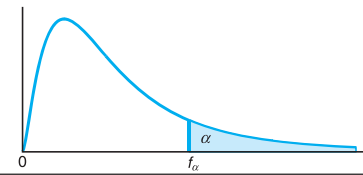
Critical Values of the Chi-Squared Distribution

| <i>v</i>  | $\alpha$             |                      |                      |                      |         |        |        |        |        |        |
|-----------|----------------------|----------------------|----------------------|----------------------|---------|--------|--------|--------|--------|--------|
|           | 0.995                | 0.99                 | 0.98                 | 0.975                | 0.95    | 0.90   | 0.80   | 0.75   | 0.70   | 0.50   |
| <b>1</b>  | 0.0 <sup>4</sup> 393 | 0.0 <sup>3</sup> 157 | 0.0 <sup>3</sup> 628 | 0.0 <sup>3</sup> 982 | 0.00393 | 0.0158 | 0.0642 | 0.102  | 0.148  | 0.455  |
| <b>2</b>  | 0.0100               | 0.0201               | 0.0404               | 0.0506               | 0.103   | 0.211  | 0.446  | 0.575  | 0.713  | 1.386  |
| <b>3</b>  | 0.0717               | 0.115                | 0.185                | 0.216                | 0.352   | 0.584  | 1.005  | 1.213  | 1.424  | 2.366  |
| <b>4</b>  | 0.207                | 0.297                | 0.429                | 0.484                | 0.711   | 1.064  | 1.649  | 1.923  | 2.195  | 3.357  |
| <b>5</b>  | 0.412                | 0.554                | 0.752                | 0.831                | 1.145   | 1.610  | 2.343  | 2.675  | 3.000  | 4.351  |
| <b>6</b>  | 0.676                | 0.872                | 1.134                | 1.237                | 1.635   | 2.204  | 3.070  | 3.455  | 3.828  | 5.348  |
| <b>7</b>  | 0.989                | 1.239                | 1.564                | 1.690                | 2.167   | 2.833  | 3.822  | 4.255  | 4.671  | 6.346  |
| <b>8</b>  | 1.344                | 1.647                | 2.032                | 2.180                | 2.733   | 3.490  | 4.594  | 5.071  | 5.527  | 7.344  |
| <b>9</b>  | 1.735                | 2.088                | 2.532                | 2.700                | 3.325   | 4.168  | 5.380  | 5.899  | 6.393  | 8.343  |
| <b>10</b> | 2.156                | 2.558                | 3.059                | 3.247                | 3.940   | 4.865  | 6.179  | 6.737  | 7.267  | 9.342  |
| <b>11</b> | 2.603                | 3.053                | 3.609                | 3.816                | 4.575   | 5.578  | 6.989  | 7.584  | 8.148  | 10.341 |
| <b>12</b> | 3.074                | 3.571                | 4.178                | 4.404                | 5.226   | 6.304  | 7.807  | 8.438  | 9.034  | 11.340 |
| <b>13</b> | 3.565                | 4.107                | 4.765                | 5.009                | 5.892   | 7.041  | 8.634  | 9.299  | 9.926  | 12.340 |
| <b>14</b> | 4.075                | 4.660                | 5.368                | 5.629                | 6.571   | 7.790  | 9.467  | 10.165 | 10.821 | 13.339 |
| <b>15</b> | 4.601                | 5.229                | 5.985                | 6.262                | 7.261   | 8.547  | 10.307 | 11.037 | 11.721 | 14.339 |
| <b>16</b> | 5.142                | 5.812                | 6.614                | 6.908                | 7.962   | 9.312  | 11.152 | 11.912 | 12.624 | 15.338 |
| <b>17</b> | 5.697                | 6.408                | 7.255                | 7.564                | 8.672   | 10.085 | 12.002 | 12.792 | 13.531 | 16.338 |
| <b>18</b> | 6.265                | 7.015                | 7.906                | 8.231                | 9.390   | 10.865 | 12.857 | 13.675 | 14.440 | 17.338 |
| <b>19</b> | 6.844                | 7.633                | 8.567                | 8.907                | 10.117  | 11.651 | 13.716 | 14.562 | 15.352 | 18.338 |
| <b>20</b> | 7.434                | 8.260                | 9.237                | 9.591                | 10.851  | 12.443 | 14.578 | 15.452 | 16.266 | 19.337 |
| <b>21</b> | 8.034                | 8.897                | 9.915                | 10.283               | 11.591  | 13.240 | 15.445 | 16.344 | 17.182 | 20.337 |
| <b>22</b> | 8.643                | 9.542                | 10.600               | 10.982               | 12.338  | 14.041 | 16.314 | 17.240 | 18.101 | 21.337 |
| <b>23</b> | 9.260                | 10.196               | 11.293               | 11.689               | 13.091  | 14.848 | 17.187 | 18.137 | 19.021 | 22.337 |
| <b>24</b> | 9.886                | 10.856               | 11.992               | 12.401               | 13.848  | 15.659 | 18.062 | 19.037 | 19.943 | 23.337 |
| <b>25</b> | 10.520               | 11.524               | 12.697               | 13.120               | 14.611  | 16.473 | 18.940 | 19.939 | 20.867 | 24.337 |
| <b>26</b> | 11.160               | 12.198               | 13.409               | 13.844               | 15.379  | 17.292 | 19.820 | 20.843 | 21.792 | 25.336 |
| <b>27</b> | 11.808               | 12.878               | 14.125               | 14.573               | 16.151  | 18.114 | 20.703 | 21.749 | 22.719 | 26.336 |
| <b>28</b> | 12.461               | 13.565               | 14.847               | 15.308               | 16.928  | 18.939 | 21.588 | 22.657 | 23.647 | 27.336 |
| <b>29</b> | 13.121               | 14.256               | 15.574               | 16.047               | 17.708  | 19.768 | 22.475 | 23.567 | 24.577 | 28.336 |
| <b>30</b> | 13.787               | 14.953               | 16.306               | 16.791               | 18.493  | 20.599 | 23.364 | 24.478 | 25.508 | 29.336 |
| <b>40</b> | 20.707               | 22.164               | 23.838               | 24.433               | 26.509  | 29.051 | 32.345 | 33.66  | 34.872 | 39.335 |
| <b>50</b> | 27.991               | 29.707               | 31.664               | 32.357               | 34.764  | 37.689 | 41.449 | 42.942 | 44.313 | 49.335 |
| <b>60</b> | 35.534               | 37.485               | 39.699               | 40.482               | 43.188  | 46.459 | 50.641 | 52.294 | 53.809 | 59.335 |

(continued) Critical Values of the Chi-Squared Distribution

| <i>v</i>  | $\alpha$ |        |        |        |        |        |        |        |        |        |
|-----------|----------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
|           | 0.30     | 0.25   | 0.20   | 0.10   | 0.05   | 0.025  | 0.02   | 0.01   | 0.005  | 0.001  |
| <b>1</b>  | 1.074    | 1.323  | 1.642  | 2.706  | 3.841  | 5.024  | 5.412  | 6.635  | 7.879  | 10.827 |
| <b>2</b>  | 2.408    | 2.773  | 3.219  | 4.605  | 5.991  | 7.378  | 7.824  | 9.210  | 10.597 | 13.815 |
| <b>3</b>  | 3.665    | 4.108  | 4.642  | 6.251  | 7.815  | 9.348  | 9.837  | 11.345 | 12.838 | 16.266 |
| <b>4</b>  | 4.878    | 5.385  | 5.989  | 7.779  | 9.488  | 11.143 | 11.668 | 13.277 | 14.860 | 18.466 |
| <b>5</b>  | 6.064    | 6.626  | 7.289  | 9.236  | 11.070 | 12.832 | 13.388 | 15.086 | 16.750 | 20.515 |
| <b>6</b>  | 7.231    | 7.841  | 8.558  | 10.645 | 12.592 | 14.449 | 15.033 | 16.812 | 18.548 | 22.457 |
| <b>7</b>  | 8.383    | 9.037  | 9.803  | 12.017 | 14.067 | 16.013 | 16.622 | 18.475 | 20.278 | 24.321 |
| <b>8</b>  | 9.524    | 10.219 | 11.030 | 13.362 | 15.507 | 17.535 | 18.168 | 20.090 | 21.955 | 26.124 |
| <b>9</b>  | 10.656   | 11.389 | 12.242 | 14.684 | 16.919 | 19.023 | 19.679 | 21.666 | 23.589 | 27.877 |
| <b>10</b> | 11.781   | 12.549 | 13.442 | 15.987 | 18.307 | 20.483 | 21.161 | 23.209 | 25.188 | 29.588 |
| <b>11</b> | 12.899   | 13.701 | 14.631 | 17.275 | 19.675 | 21.920 | 22.618 | 24.725 | 26.757 | 31.264 |
| <b>12</b> | 14.011   | 14.845 | 15.812 | 18.549 | 21.026 | 23.337 | 24.054 | 26.217 | 28.300 | 32.909 |
| <b>13</b> | 15.119   | 15.984 | 16.985 | 19.812 | 22.362 | 24.736 | 25.471 | 27.688 | 29.819 | 34.527 |
| <b>14</b> | 16.222   | 17.117 | 18.151 | 21.064 | 23.685 | 26.119 | 26.873 | 29.141 | 31.319 | 36.124 |
| <b>15</b> | 17.322   | 18.245 | 19.311 | 22.307 | 24.996 | 27.488 | 28.259 | 30.578 | 32.801 | 37.698 |
| <b>16</b> | 18.418   | 19.369 | 20.465 | 23.542 | 26.296 | 28.845 | 29.633 | 32.000 | 34.267 | 39.252 |
| <b>17</b> | 19.511   | 20.489 | 21.615 | 24.769 | 27.587 | 30.191 | 30.995 | 33.409 | 35.718 | 40.791 |
| <b>18</b> | 20.601   | 21.605 | 22.760 | 25.989 | 28.869 | 31.526 | 32.346 | 34.805 | 37.156 | 42.312 |
| <b>19</b> | 21.689   | 22.718 | 23.900 | 27.204 | 30.144 | 32.852 | 33.687 | 36.191 | 38.582 | 43.819 |
| <b>20</b> | 22.775   | 23.828 | 25.038 | 28.412 | 31.410 | 34.170 | 35.020 | 37.566 | 39.997 | 45.314 |
| <b>21</b> | 23.858   | 24.935 | 26.171 | 29.615 | 32.671 | 35.479 | 36.343 | 38.932 | 41.401 | 46.796 |
| <b>22</b> | 24.939   | 26.039 | 27.301 | 30.813 | 33.924 | 36.781 | 37.659 | 40.289 | 42.796 | 48.268 |
| <b>23</b> | 26.018   | 27.141 | 28.429 | 32.007 | 35.172 | 38.076 | 38.968 | 41.638 | 44.181 | 49.728 |
| <b>24</b> | 27.096   | 28.241 | 29.553 | 33.196 | 36.415 | 39.364 | 40.270 | 42.980 | 45.558 | 51.179 |
| <b>25</b> | 28.172   | 29.339 | 30.675 | 34.382 | 37.652 | 40.646 | 41.566 | 44.314 | 46.928 | 52.619 |
| <b>26</b> | 29.246   | 30.435 | 31.795 | 35.563 | 38.885 | 41.923 | 42.856 | 45.642 | 48.290 | 54.051 |
| <b>27</b> | 30.319   | 31.528 | 32.912 | 36.741 | 40.113 | 43.195 | 44.140 | 46.963 | 49.645 | 55.475 |
| <b>28</b> | 31.391   | 32.620 | 34.027 | 37.916 | 41.337 | 44.461 | 45.419 | 48.278 | 50.994 | 56.892 |
| <b>29</b> | 32.461   | 33.711 | 35.139 | 39.087 | 42.557 | 45.722 | 46.693 | 49.588 | 52.335 | 58.301 |
| <b>30</b> | 33.530   | 34.800 | 36.250 | 40.256 | 43.773 | 46.979 | 47.962 | 50.892 | 53.672 | 59.702 |
| <b>40</b> | 44.165   | 45.616 | 47.269 | 51.805 | 55.758 | 59.342 | 60.436 | 63.691 | 66.766 | 73.403 |
| <b>50</b> | 54.723   | 56.334 | 58.164 | 63.167 | 67.505 | 71.420 | 72.613 | 76.154 | 79.490 | 86.660 |
| <b>60</b> | 65.226   | 66.981 | 68.972 | 74.397 | 79.082 | 83.298 | 84.58  | 88.379 | 91.952 | 99.608 |

Critical Values of the  $F$ -Distribution



| $f_{0.05}(v_1, v_2)$ |        |        |        |        |        |        |        |        |        |
|----------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| $v_2$                | $v_1$  |        |        |        |        |        |        |        |        |
|                      | 1      | 2      | 3      | 4      | 5      | 6      | 7      | 8      | 9      |
| 1                    | 161.45 | 199.50 | 215.71 | 224.58 | 230.16 | 233.99 | 236.77 | 238.88 | 240.54 |
| 2                    | 18.51  | 19.00  | 19.16  | 19.25  | 19.30  | 19.33  | 19.35  | 19.37  | 19.38  |
| 3                    | 10.13  | 9.55   | 9.28   | 9.12   | 9.01   | 8.94   | 8.89   | 8.85   | 8.81   |
| 4                    | 7.71   | 6.94   | 6.59   | 6.39   | 6.26   | 6.16   | 6.09   | 6.04   | 6.00   |
| 5                    | 6.61   | 5.79   | 5.41   | 5.19   | 5.05   | 4.95   | 4.88   | 4.82   | 4.77   |
| 6                    | 5.99   | 5.14   | 4.76   | 4.53   | 4.39   | 4.28   | 4.21   | 4.15   | 4.10   |
| 7                    | 5.59   | 4.74   | 4.35   | 4.12   | 3.97   | 3.87   | 3.79   | 3.73   | 3.68   |
| 8                    | 5.32   | 4.46   | 4.07   | 3.84   | 3.69   | 3.58   | 3.50   | 3.44   | 3.39   |
| 9                    | 5.12   | 4.26   | 3.86   | 3.63   | 3.48   | 3.37   | 3.29   | 3.23   | 3.18   |
| 10                   | 4.96   | 4.10   | 3.71   | 3.48   | 3.33   | 3.22   | 3.14   | 3.07   | 3.02   |
| 11                   | 4.84   | 3.98   | 3.59   | 3.36   | 3.20   | 3.09   | 3.01   | 2.95   | 2.90   |
| 12                   | 4.75   | 3.89   | 3.49   | 3.26   | 3.11   | 3.00   | 2.91   | 2.85   | 2.80   |
| 13                   | 4.67   | 3.81   | 3.41   | 3.18   | 3.03   | 2.92   | 2.83   | 2.77   | 2.71   |
| 14                   | 4.60   | 3.74   | 3.34   | 3.11   | 2.96   | 2.85   | 2.76   | 2.70   | 2.65   |
| 15                   | 4.54   | 3.68   | 3.29   | 3.06   | 2.90   | 2.79   | 2.71   | 2.64   | 2.59   |
| 16                   | 4.49   | 3.63   | 3.24   | 3.01   | 2.85   | 2.74   | 2.66   | 2.59   | 2.54   |
| 17                   | 4.45   | 3.59   | 3.20   | 2.96   | 2.81   | 2.70   | 2.61   | 2.55   | 2.49   |
| 18                   | 4.41   | 3.55   | 3.16   | 2.93   | 2.77   | 2.66   | 2.58   | 2.51   | 2.46   |
| 19                   | 4.38   | 3.52   | 3.13   | 2.90   | 2.74   | 2.63   | 2.54   | 2.48   | 2.42   |
| 20                   | 4.35   | 3.49   | 3.10   | 2.87   | 2.71   | 2.60   | 2.51   | 2.45   | 2.39   |
| 21                   | 4.32   | 3.47   | 3.07   | 2.84   | 2.68   | 2.57   | 2.49   | 2.42   | 2.37   |
| 22                   | 4.30   | 3.44   | 3.05   | 2.82   | 2.66   | 2.55   | 2.46   | 2.40   | 2.34   |
| 23                   | 4.28   | 3.42   | 3.03   | 2.80   | 2.64   | 2.53   | 2.44   | 2.37   | 2.32   |
| 24                   | 4.26   | 3.40   | 3.01   | 2.78   | 2.62   | 2.51   | 2.42   | 2.36   | 2.30   |
| 25                   | 4.24   | 3.39   | 2.99   | 2.76   | 2.60   | 2.49   | 2.40   | 2.34   | 2.28   |
| 26                   | 4.23   | 3.37   | 2.98   | 2.74   | 2.59   | 2.47   | 2.39   | 2.32   | 2.27   |
| 27                   | 4.21   | 3.35   | 2.96   | 2.73   | 2.57   | 2.46   | 2.37   | 2.31   | 2.25   |
| 28                   | 4.20   | 3.34   | 2.95   | 2.71   | 2.56   | 2.45   | 2.36   | 2.29   | 2.24   |
| 29                   | 4.18   | 3.33   | 2.93   | 2.70   | 2.55   | 2.43   | 2.35   | 2.28   | 2.22   |
| 30                   | 4.17   | 3.32   | 2.92   | 2.69   | 2.53   | 2.42   | 2.33   | 2.27   | 2.21   |
| 40                   | 4.08   | 3.23   | 2.84   | 2.61   | 2.45   | 2.34   | 2.25   | 2.18   | 2.12   |
| 60                   | 4.00   | 3.15   | 2.76   | 2.53   | 2.37   | 2.25   | 2.17   | 2.10   | 2.04   |
| 120                  | 3.92   | 3.07   | 2.68   | 2.45   | 2.29   | 2.18   | 2.09   | 2.02   | 1.96   |
| $\infty$             | 3.84   | 3.00   | 2.60   | 2.37   | 2.21   | 2.10   | 2.01   | 1.94   | 1.88   |

(continued) Critical Values of the  $F$ -Distribution

| $v_2$    | $f_{0.05}(v_1, v_2)$ |        |        |        |        |        |        |        |        |          |
|----------|----------------------|--------|--------|--------|--------|--------|--------|--------|--------|----------|
|          | $v_1$                |        |        |        |        |        |        |        |        |          |
|          | 10                   | 12     | 15     | 20     | 24     | 30     | 40     | 60     | 120    | $\infty$ |
| 1        | 241.88               | 243.91 | 245.95 | 248.01 | 249.05 | 250.10 | 251.14 | 252.20 | 253.25 | 254.31   |
| 2        | 19.40                | 19.41  | 19.43  | 19.45  | 19.45  | 19.46  | 19.47  | 19.48  | 19.49  | 19.50    |
| 3        | 8.79                 | 8.74   | 8.70   | 8.66   | 8.64   | 8.62   | 8.59   | 8.57   | 8.55   | 8.53     |
| 4        | 5.96                 | 5.91   | 5.86   | 5.80   | 5.77   | 5.75   | 5.72   | 5.69   | 5.66   | 5.63     |
| 5        | 4.74                 | 4.68   | 4.62   | 4.56   | 4.53   | 4.50   | 4.46   | 4.43   | 4.40   | 4.36     |
| 6        | 4.06                 | 4.00   | 3.94   | 3.87   | 3.84   | 3.81   | 3.77   | 3.74   | 3.70   | 3.67     |
| 7        | 3.64                 | 3.57   | 3.51   | 3.44   | 3.41   | 3.38   | 3.34   | 3.30   | 3.27   | 3.23     |
| 8        | 3.35                 | 3.28   | 3.22   | 3.15   | 3.12   | 3.08   | 3.04   | 3.01   | 2.97   | 2.93     |
| 9        | 3.14                 | 3.07   | 3.01   | 2.94   | 2.90   | 2.86   | 2.83   | 2.79   | 2.75   | 2.71     |
| 10       | 2.98                 | 2.91   | 2.85   | 2.77   | 2.74   | 2.70   | 2.66   | 2.62   | 2.58   | 2.54     |
| 11       | 2.85                 | 2.79   | 2.72   | 2.65   | 2.61   | 2.57   | 2.53   | 2.49   | 2.45   | 2.40     |
| 12       | 2.75                 | 2.69   | 2.62   | 2.54   | 2.51   | 2.47   | 2.43   | 2.38   | 2.34   | 2.30     |
| 13       | 2.67                 | 2.60   | 2.53   | 2.46   | 2.42   | 2.38   | 2.34   | 2.30   | 2.25   | 2.21     |
| 14       | 2.60                 | 2.53   | 2.46   | 2.39   | 2.35   | 2.31   | 2.27   | 2.22   | 2.18   | 2.13     |
| 15       | 2.54                 | 2.48   | 2.40   | 2.33   | 2.29   | 2.25   | 2.20   | 2.16   | 2.11   | 2.07     |
| 16       | 2.49                 | 2.42   | 2.35   | 2.28   | 2.24   | 2.19   | 2.15   | 2.11   | 2.06   | 2.01     |
| 17       | 2.45                 | 2.38   | 2.31   | 2.23   | 2.19   | 2.15   | 2.10   | 2.06   | 2.01   | 1.96     |
| 18       | 2.41                 | 2.34   | 2.27   | 2.19   | 2.15   | 2.11   | 2.06   | 2.02   | 1.97   | 1.92     |
| 19       | 2.38                 | 2.31   | 2.23   | 2.16   | 2.11   | 2.07   | 2.03   | 1.98   | 1.93   | 1.88     |
| 20       | 2.35                 | 2.28   | 2.20   | 2.12   | 2.08   | 2.04   | 1.99   | 1.95   | 1.90   | 1.84     |
| 21       | 2.32                 | 2.25   | 2.18   | 2.10   | 2.05   | 2.01   | 1.96   | 1.92   | 1.87   | 1.81     |
| 22       | 2.30                 | 2.23   | 2.15   | 2.07   | 2.03   | 1.98   | 1.94   | 1.89   | 1.84   | 1.78     |
| 23       | 2.27                 | 2.20   | 2.13   | 2.05   | 2.01   | 1.96   | 1.91   | 1.86   | 1.81   | 1.76     |
| 24       | 2.25                 | 2.18   | 2.11   | 2.03   | 1.98   | 1.94   | 1.89   | 1.84   | 1.79   | 1.73     |
| 25       | 2.24                 | 2.16   | 2.09   | 2.01   | 1.96   | 1.92   | 1.87   | 1.82   | 1.77   | 1.71     |
| 26       | 2.22                 | 2.15   | 2.07   | 1.99   | 1.95   | 1.90   | 1.85   | 1.80   | 1.75   | 1.69     |
| 27       | 2.20                 | 2.13   | 2.06   | 1.97   | 1.93   | 1.88   | 1.84   | 1.79   | 1.73   | 1.67     |
| 28       | 2.19                 | 2.12   | 2.04   | 1.96   | 1.91   | 1.87   | 1.82   | 1.77   | 1.71   | 1.65     |
| 29       | 2.18                 | 2.10   | 2.03   | 1.94   | 1.90   | 1.85   | 1.81   | 1.75   | 1.70   | 1.64     |
| 30       | 2.16                 | 2.09   | 2.01   | 1.93   | 1.89   | 1.84   | 1.79   | 1.74   | 1.68   | 1.62     |
| 40       | 2.08                 | 2.00   | 1.92   | 1.84   | 1.79   | 1.74   | 1.69   | 1.64   | 1.58   | 1.51     |
| 60       | 1.99                 | 1.92   | 1.84   | 1.75   | 1.70   | 1.65   | 1.59   | 1.53   | 1.47   | 1.39     |
| 120      | 1.91                 | 1.83   | 1.75   | 1.66   | 1.61   | 1.55   | 1.50   | 1.43   | 1.35   | 1.25     |
| $\infty$ | 1.83                 | 1.75   | 1.67   | 1.57   | 1.52   | 1.46   | 1.39   | 1.32   | 1.22   | 1.00     |

(continued) Critical Values of the  $F$ -Distribution

| $v_2$    | $f_{0.01}(v_1, v_2)$ |         |         |         |         |         |         |         |         |
|----------|----------------------|---------|---------|---------|---------|---------|---------|---------|---------|
|          | $v_1$                |         |         |         |         |         |         |         |         |
|          | 1                    | 2       | 3       | 4       | 5       | 6       | 7       | 8       | 9       |
| 1        | 4052.18              | 4999.50 | 5403.35 | 5624.58 | 5763.65 | 5858.99 | 5928.36 | 5981.07 | 6022.47 |
| 2        | 98.50                | 99.00   | 99.17   | 99.25   | 99.30   | 99.33   | 99.36   | 99.37   | 99.39   |
| 3        | 34.12                | 30.82   | 29.46   | 28.71   | 28.24   | 27.91   | 27.67   | 27.49   | 27.35   |
| 4        | 21.20                | 18.00   | 16.69   | 15.98   | 15.52   | 15.21   | 14.98   | 14.80   | 14.66   |
| 5        | 16.26                | 13.27   | 12.06   | 11.39   | 10.97   | 10.67   | 10.46   | 10.29   | 10.16   |
| 6        | 13.75                | 10.92   | 9.78    | 9.15    | 8.75    | 8.47    | 8.26    | 8.10    | 7.98    |
| 7        | 12.25                | 9.55    | 8.45    | 7.85    | 7.46    | 7.19    | 6.99    | 6.84    | 6.72    |
| 8        | 11.26                | 8.65    | 7.59    | 7.01    | 6.63    | 6.37    | 6.18    | 6.03    | 5.91    |
| 9        | 10.56                | 8.02    | 6.99    | 6.42    | 6.06    | 5.80    | 5.61    | 5.47    | 5.35    |
| 10       | 10.04                | 7.56    | 6.55    | 5.99    | 5.64    | 5.39    | 5.20    | 5.06    | 4.94    |
| 11       | 9.65                 | 7.21    | 6.22    | 5.67    | 5.32    | 5.07    | 4.89    | 4.74    | 4.63    |
| 12       | 9.33                 | 6.93    | 5.95    | 5.41    | 5.06    | 4.82    | 4.64    | 4.50    | 4.39    |
| 13       | 9.07                 | 6.70    | 5.74    | 5.21    | 4.86    | 4.62    | 4.44    | 4.30    | 4.19    |
| 14       | 8.86                 | 6.51    | 5.56    | 5.04    | 4.69    | 4.46    | 4.28    | 4.14    | 4.03    |
| 15       | 8.68                 | 6.36    | 5.42    | 4.89    | 4.56    | 4.32    | 4.14    | 4.00    | 3.89    |
| 16       | 8.53                 | 6.23    | 5.29    | 4.77    | 4.44    | 4.20    | 4.03    | 3.89    | 3.78    |
| 17       | 8.40                 | 6.11    | 5.18    | 4.67    | 4.34    | 4.10    | 3.93    | 3.79    | 3.68    |
| 18       | 8.29                 | 6.01    | 5.09    | 4.58    | 4.25    | 4.01    | 3.84    | 3.71    | 3.60    |
| 19       | 8.18                 | 5.93    | 5.01    | 4.50    | 4.17    | 3.94    | 3.77    | 3.63    | 3.52    |
| 20       | 8.10                 | 5.85    | 4.94    | 4.43    | 4.10    | 3.87    | 3.70    | 3.56    | 3.46    |
| 21       | 8.02                 | 5.78    | 4.87    | 4.37    | 4.04    | 3.81    | 3.64    | 3.51    | 3.40    |
| 22       | 7.95                 | 5.72    | 4.82    | 4.31    | 3.99    | 3.76    | 3.59    | 3.45    | 3.35    |
| 23       | 7.88                 | 5.66    | 4.76    | 4.26    | 3.94    | 3.71    | 3.54    | 3.41    | 3.30    |
| 24       | 7.82                 | 5.61    | 4.72    | 4.22    | 3.90    | 3.67    | 3.50    | 3.36    | 3.26    |
| 25       | 7.77                 | 5.57    | 4.68    | 4.18    | 3.85    | 3.63    | 3.46    | 3.32    | 3.22    |
| 26       | 7.72                 | 5.53    | 4.64    | 4.14    | 3.82    | 3.59    | 3.42    | 3.29    | 3.18    |
| 27       | 7.68                 | 5.49    | 4.60    | 4.11    | 3.78    | 3.56    | 3.39    | 3.26    | 3.15    |
| 28       | 7.64                 | 5.45    | 4.57    | 4.07    | 3.75    | 3.53    | 3.36    | 3.23    | 3.12    |
| 29       | 7.60                 | 5.42    | 4.54    | 4.04    | 3.73    | 3.50    | 3.33    | 3.20    | 3.09    |
| 30       | 7.56                 | 5.39    | 4.51    | 4.02    | 3.70    | 3.47    | 3.30    | 3.17    | 3.07    |
| 40       | 7.31                 | 5.18    | 4.31    | 3.83    | 3.51    | 3.29    | 3.12    | 2.99    | 2.89    |
| 60       | 7.08                 | 4.98    | 4.13    | 3.65    | 3.34    | 3.12    | 2.95    | 2.82    | 2.72    |
| 120      | 6.85                 | 4.79    | 3.95    | 3.48    | 3.17    | 2.96    | 2.79    | 2.66    | 2.56    |
| $\infty$ | 6.63                 | 4.61    | 3.78    | 3.32    | 3.02    | 2.80    | 2.64    | 2.51    | 2.41    |

(continued) Critical Values of the  $F$ -Distribution

| $v_2$    | $f_{0.01}(v_1, v_2)$ |         |         |         |         |         |         |         |         |          |
|----------|----------------------|---------|---------|---------|---------|---------|---------|---------|---------|----------|
|          | $v_1$                |         |         |         |         |         |         |         |         |          |
|          | 10                   | 12      | 15      | 20      | 24      | 30      | 40      | 60      | 120     | $\infty$ |
| 1        | 6055.85              | 6106.32 | 6157.28 | 6208.73 | 6234.63 | 6260.65 | 6286.78 | 6313.03 | 6339.39 | 6365.86  |
| 2        | 99.40                | 99.42   | 99.43   | 99.45   | 99.46   | 99.47   | 99.47   | 99.48   | 99.49   | 99.50    |
| 3        | 27.23                | 27.05   | 26.87   | 26.69   | 26.60   | 26.50   | 26.41   | 26.32   | 26.22   | 26.13    |
| 4        | 14.55                | 14.37   | 14.20   | 14.02   | 13.93   | 13.84   | 13.75   | 13.65   | 13.56   | 13.46    |
| 5        | 10.05                | 9.89    | 9.72    | 9.55    | 9.47    | 9.38    | 9.29    | 9.20    | 9.11    | 9.02     |
| 6        | 7.87                 | 7.72    | 7.56    | 7.40    | 7.31    | 7.23    | 7.14    | 7.06    | 6.97    | 6.88     |
| 7        | 6.62                 | 6.47    | 6.31    | 6.16    | 6.07    | 5.99    | 5.91    | 5.82    | 5.74    | 5.65     |
| 8        | 5.81                 | 5.67    | 5.52    | 5.36    | 5.28    | 5.20    | 5.12    | 5.03    | 4.95    | 4.86     |
| 9        | 5.26                 | 5.11    | 4.96    | 4.81    | 4.73    | 4.65    | 4.57    | 4.48    | 4.40    | 4.31     |
| 10       | 4.85                 | 4.71    | 4.56    | 4.41    | 4.33    | 4.25    | 4.17    | 4.08    | 4.00    | 3.91     |
| 11       | 4.54                 | 4.40    | 4.25    | 4.10    | 4.02    | 3.94    | 3.86    | 3.78    | 3.69    | 3.60     |
| 12       | 4.30                 | 4.16    | 4.01    | 3.86    | 3.78    | 3.70    | 3.62    | 3.54    | 3.45    | 3.36     |
| 13       | 4.10                 | 3.96    | 3.82    | 3.66    | 3.59    | 3.51    | 3.43    | 3.34    | 3.25    | 3.17     |
| 14       | 3.94                 | 3.80    | 3.66    | 3.51    | 3.43    | 3.35    | 3.27    | 3.18    | 3.09    | 3.00     |
| 15       | 3.80                 | 3.67    | 3.52    | 3.37    | 3.29    | 3.21    | 3.13    | 3.05    | 2.96    | 2.87     |
| 16       | 3.69                 | 3.55    | 3.41    | 3.26    | 3.18    | 3.10    | 3.02    | 2.93    | 2.84    | 2.75     |
| 17       | 3.59                 | 3.46    | 3.31    | 3.16    | 3.08    | 3.00    | 2.92    | 2.83    | 2.75    | 2.65     |
| 18       | 3.51                 | 3.37    | 3.23    | 3.08    | 3.00    | 2.92    | 2.84    | 2.75    | 2.66    | 2.57     |
| 19       | 3.43                 | 3.30    | 3.15    | 3.00    | 2.92    | 2.84    | 2.76    | 2.67    | 2.58    | 2.49     |
| 20       | 3.37                 | 3.23    | 3.09    | 2.94    | 2.86    | 2.78    | 2.69    | 2.61    | 2.52    | 2.42     |
| 21       | 3.31                 | 3.17    | 3.03    | 2.88    | 2.80    | 2.72    | 2.64    | 2.55    | 2.46    | 2.36     |
| 22       | 3.26                 | 3.12    | 2.98    | 2.83    | 2.75    | 2.67    | 2.58    | 2.50    | 2.40    | 2.31     |
| 23       | 3.21                 | 3.07    | 2.93    | 2.78    | 2.70    | 2.62    | 2.54    | 2.45    | 2.35    | 2.26     |
| 24       | 3.17                 | 3.03    | 2.89    | 2.74    | 2.66    | 2.58    | 2.49    | 2.40    | 2.31    | 2.21     |
| 25       | 3.13                 | 2.99    | 2.85    | 2.70    | 2.62    | 2.54    | 2.45    | 2.36    | 2.27    | 2.17     |
| 26       | 3.09                 | 2.96    | 2.81    | 2.66    | 2.58    | 2.50    | 2.42    | 2.33    | 2.23    | 2.13     |
| 27       | 3.06                 | 2.93    | 2.78    | 2.63    | 2.55    | 2.47    | 2.38    | 2.29    | 2.20    | 2.10     |
| 28       | 3.03                 | 2.90    | 2.75    | 2.60    | 2.52    | 2.44    | 2.35    | 2.26    | 2.17    | 2.06     |
| 29       | 3.00                 | 2.87    | 2.73    | 2.57    | 2.49    | 2.41    | 2.33    | 2.23    | 2.14    | 2.03     |
| 30       | 2.98                 | 2.84    | 2.70    | 2.55    | 2.47    | 2.39    | 2.30    | 2.21    | 2.11    | 2.01     |
| 40       | 2.80                 | 2.66    | 2.52    | 2.37    | 2.29    | 2.20    | 2.11    | 2.02    | 1.92    | 1.80     |
| 60       | 2.63                 | 2.50    | 2.35    | 2.20    | 2.12    | 2.03    | 1.94    | 1.84    | 1.73    | 1.60     |
| 120      | 2.47                 | 2.34    | 2.19    | 2.03    | 1.95    | 1.86    | 1.76    | 1.66    | 1.53    | 1.38     |
| $\infty$ | 2.32                 | 2.18    | 2.04    | 1.88    | 1.79    | 1.70    | 1.59    | 1.47    | 1.32    | 1.00     |